

Chapter 53. Fighting the Ventilator

Martin J. Tobin; Amal Jubran; Franco Laghi

Fighting the Ventilator: Introduction

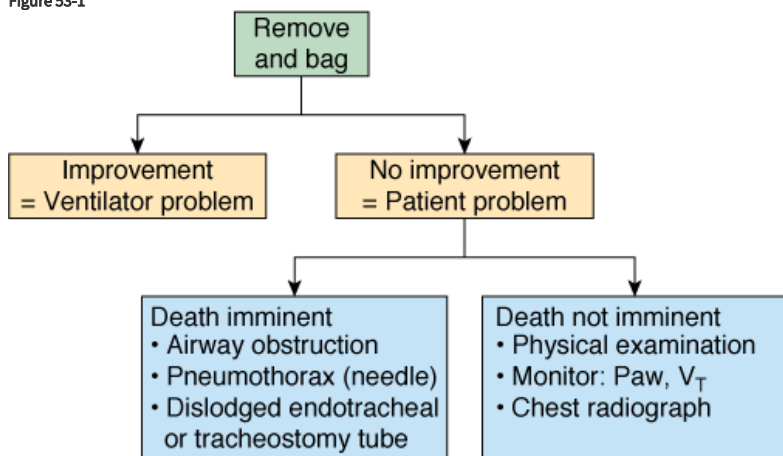
A patient is connected to the ventilator. His eyes are closed and he appears calm. The ventilator is making soft rhythmic noises, and the patient's chest is expanding and receding in unison with the ventilator. A sweep of signals is gently traversing the monitor screen.

This peaceful scene erupts suddenly. The patient bolts upright. His eyelids retract. His nostrils flare. Sweat drips from his brow. His skin turns blue. His sternocleidomastoids contract vigorously. His rib spaces retract. One or more alarms sound loudly.

A patient fighting (or bucking) the ventilator is frightening not only to the patient but to staff. If the physician cannot find the source of the problem and fix it, the patient may die in minutes. The physician must immediately diagnose and manage the problem, and do both concurrently. The physician quickly scans the monitors for clues. Sometimes the problem is immediately spotted and solved, such as disconnected ventilator tubing.

If the cause is not immediately obvious, the physician's primary responsibility is to ensure adequate ventilation. This requirement takes precedence over diagnosis. After disconnecting the patient from the ventilator, the physician (or staff) starts to ventilate the patient manually with a self-inflating bag containing 100% oxygen. This step is both therapeutic and diagnostic (Fig. 53-1).^{1,2} If the distress resolves, it indicates that the problem originated in the ventilator. If the distress continues, it indicates the problem is within the patient.

Figure 53-1



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Removing the patient from the ventilator and providing manual ventilation with a bag (100% oxygen) is both therapeutic and diagnostic.

Patient improvement indicates that the ventilator is the cause of the distress. Lack of improvement indicates that the problem is within the patient. If death appears imminent, the physician rapidly checks for airway obstruction (by passing a suction catheter), a dislodged endotracheal or tracheostomy tube, and a pneumothorax (if deemed likely, a small-gauge needle is inserted); depending on the clinical picture, other life-threatening problems, such as aortic dissection, should be considered. If death is not imminent, the physician undertakes a more detailed physical examination and assessment of monitored variables. A chest radiograph may also be obtained. *Paw*, airway pressure; *V_T*, tidal volume.

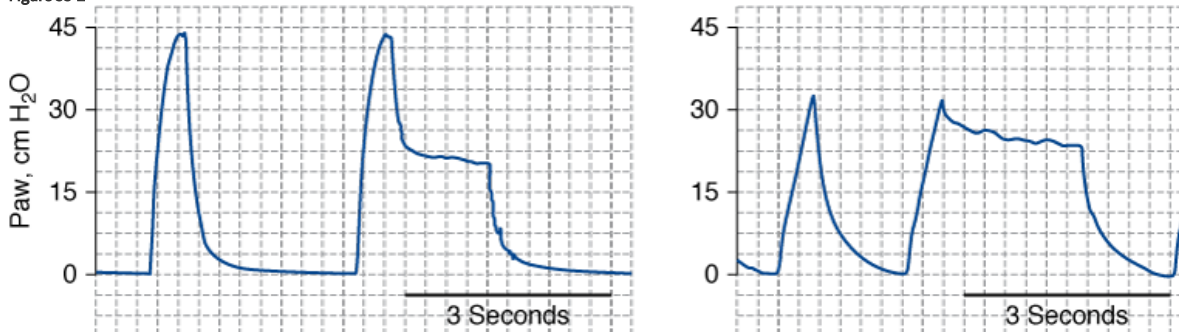
Initial Assessment

Where a physician begins the assessment varies with the particulars of the patient's presentation. The following sequence is not appropriate for all patients.

Because hypoxia can be rapidly lethal, the pulse oximeter reading is noted. Although several factors can give rise to erroneous readings, the displayed saturation generally bears a close relationship to the [oxygen](#) saturation on an arterial blood-gas test.³

If the high-airway-pressure alarm is sounding, the physician should, if possible, perform an end-inspiratory occlusion maneuver to measure the plateau pressure ([Fig. 53-2](#)). An increase in peak airway pressure without a proportional increase in plateau pressure indicates narrowing of the airway.⁴ Narrowing may arise from bronchoconstriction, an increase in secretions, a collection of fluid in the inspiratory limb of the ventilator circuit, kinking of the tubing, a foreign body, or herniation of the cuff of the endotracheal tube over its distal tip.

Figure 53-2



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End-inspiratory occlusion maneuver during passive ventilation in two patients. In each patient, the second displayed breath was occluded for 2 seconds or less—a period that was not sufficient to achieve a complete plateau in either patient. *Left panel.* Airway pressure (P_{aw}) drops by 24 cm H₂O during the end-inspiratory occlusion maneuver, indicating the presence of increased airflow resistance. *Right panel.* P_{aw} drops by 5 cm H₂O during the end-inspiratory occlusion maneuver, indicating decreased respiratory compliance consequent to noncardiogenic pulmonary edema secondary to sepsis.

An increase in peak airway pressure accompanied by a proportional increase in plateau pressure indicates a decrease in thoracic compliance, which may arise within the lung or from extrinsic compression. A decrease in lung compliance may arise from pulmonary edema, early stages of the acute respiratory distress syndrome, dynamic hyperinflation associated with intrinsic positive end-expiratory pressure (PEEPi), and atelectasis. Extrinsic compression may be caused by rapid collection of pleural fluid (hemothorax, parapneumonic effusion, or transudative effusion) or a pneumothorax. If a tension pneumothorax is suspected and death appears imminent, the physician should insert a 14- to 16-gauge needle into the second intercostal space. If a gush of air is heard, a thoracostomy tube is inserted. Gastric distension, ileus, ascites, or extensive eschar formation can also cause a decrease in thoracic compliance.

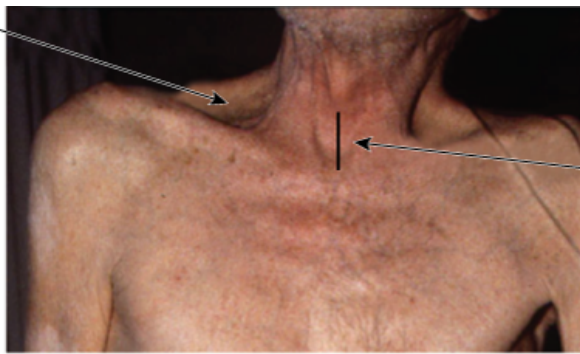
A major obstruction of the airway will commonly cause [oxygen](#) desaturation and an increase in peak airway pressure. If obstruction is suspected, the clinician should pass a suction catheter through the airway to determine if the airway is patent and to remove secretions or other material that is causing a blockage.

The alarm may signal a low exhaled tidal volume (V_T). This can arise because of movement of the endotracheal tube into the hypopharynx or esophagus. It can also arise because of a disconnection in the ventilator circuit or a leak in the system. Many conditions can cause an increase in respiratory rate. Accordingly, a high rate is a very sensitive marker of an important change in a patient's condition. Because it is influenced by so many factors, it is not helpful in directing the clinician to the cause of the new change.

The patient's pattern of breathing may signal a marked increase in work of breathing. Important signs include tracheal tug (how much the larynx descends with each inspiration); palpable activity of the sternocleidomastoid muscles; recession in the suprasternal notch, supraclavicular space, and intercostal spaces; and paradoxical movement of the rib cage and abdomen ([Fig. 53-3](#)).

Figure 53-3

Recession of supraclavicular fossa



Decrease in cricosternal distance

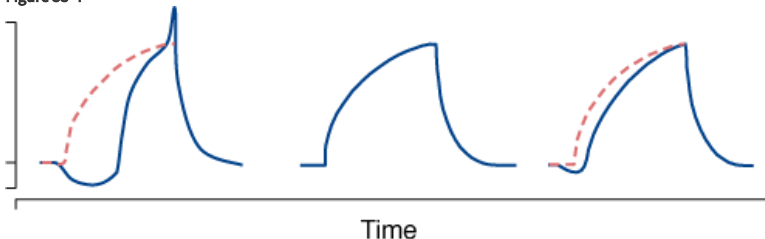
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Examination of the neck in a patient with acute respiratory distress. Inspection reveals recession of the supraclavicular fossa, recession of the suprasternal notch, increased tracheal tug (not visible on the photograph), and prominent sternocleidomastoid muscles. Palpation confirms phasic activity of the sternocleidomastoid muscles, tracheal tug, and decreased cricosternal distance, indicative of hyperinflation.

Auscultation of the chest may suggest a pneumothorax or movement of the endotracheal tube into the right main-stem bronchus. Cardiac auscultation may reveal the Hamman crunch, signaling the presence of pneumomediastinum. A new murmur may signal the development of a ventricular septal defect or other cardiac source of acute distress.

The pressure and flow tracings on the ventilator screen may provide clues to the sudden change in patient status. The contour of the airway pressure tracing may exhibit excessive scooping, indicating increased inspiratory effort; this pattern is seen when the delivered inspiratory flow no longer meets patient demand (Fig. 53-4).^{5,6} The flow tracing may not return to the zero-flow line in the period immediately before the next inspiration, indicating PEEP.⁴ The flow tracing may exhibit a sawtooth pattern, indicating excessive secretions.⁷

Figure 53-4



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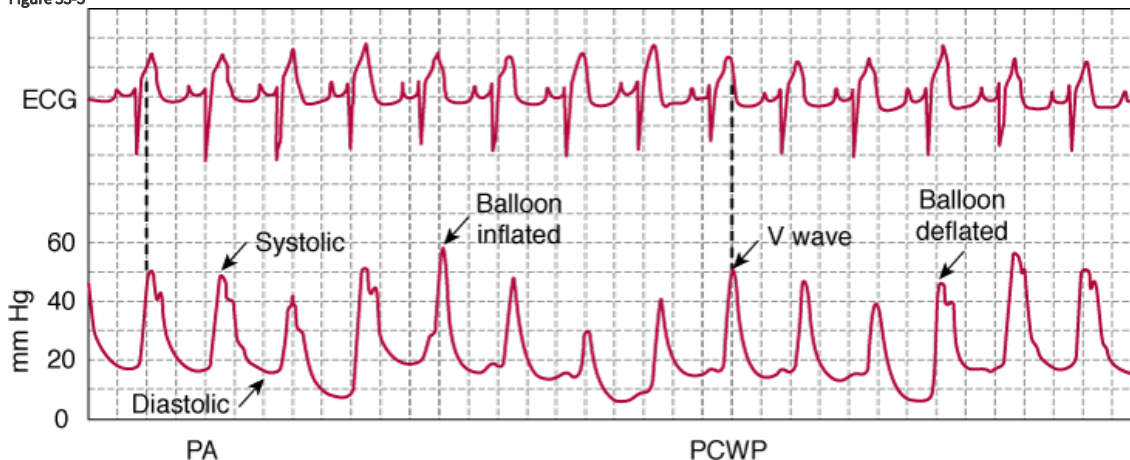
Airway pressure waveforms recorded in a patient shortly after the initiation of mechanical ventilation (*left*), in a patient making no respiratory effort (controlled mechanical ventilation, *middle*), and in a patient receiving an appropriate level of assist-control mechanical ventilation (*right*). The *dashed lines* on the left and right panels reproduce the tracing achieved by passive, controlled mechanical ventilation as occurs in a patient receiving neuromuscular blocking agents. The *left waveform* depicts a patient in respiratory distress who has excessive work of breathing; this can be inferred from the initial concavity, which results from vigorous inspiratory effort, and the spike at the end of ventilator assistance, which is the result of expiratory muscle recruitment. The *middle waveform* depicts a patient making no respiratory effort. The *right waveform* depicts a patient performing an appropriate amount of respiratory work: The small downward dip at the start of the breath indicates the small inspiratory effort required to trigger the ventilator, and the distance between the *solid line* (actual airway pressure) and *dashed line* (expected tracing during controlled ventilation, as in the middle panel) is proportional to the amount of work performed by the patient's inspiratory muscles while the ventilator is providing assistance. The patient in the right panel is performing much more respiratory work than the patient in the middle panel and much less work than the patient in the left panel. (Used, with permission, from Tobin et al.⁶)

A decrease in end-tidal carbon dioxide tension (P_{CO_2}) can be helpful. A sudden decrease in end-tidal P_{CO_2} may arise with pulmonary embolism or air embolism. Esophageal intubation also causes end-tidal P_{CO_2} to fall to zero. Unfortunately, end-tidal CO_2 readings do not reliably reflect arterial CO_2 tension (P_{CO_2}) in patients with underlying lung disease.

The blood pressure reading may signal a hypertensive crisis. The differential diagnosis of a fall in arterial pressure is wide, and influenced by clinical context. In a patient with a myocardial infarction, a fall in arterial pressure may signal the onset of cardiogenic shock. A pulmonary artery catheter can provide important clues. A new onset of large V (ventricular) waves in the pulmonary artery wedge tracing suggests the

development of acute mitral regurgitation secondary to ruptured chordae tendineae cordis (Fig. 53-5).⁸ The electrocardiogram (ECG) may reveal arrhythmias or ST-segment elevation. In contrast to the average patient, communication with an intubated patient is very difficult. A nonintubated patient who develops acute distress consequent to an acute myocardial infarction is able to relay the onset of crushing chest pain to staff. The presence of a tracheal tube prohibits speech. Nevertheless, clinicians can obtain a considerable amount of information about the nature of symptoms. Nurses are frequently more patient and skilled than physicians in achieving an ongoing dialogue with a patient.

Figure 53-5



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Simultaneous recordings of electrocardiogram (ECG, *top*) and pulmonary artery (PA) pressure (*bottom*) in a patient with acute mitral insufficiency. Following inflation of the balloon, giant V waves are seen in the pulmonary capillary wedge pressure (PCWP) tracing. Before balloon inflation and after balloon deflation, the PA tracing has a characteristic bifid appearance secondary to both a PA systolic wave and a V wave. Note that the V wave occurs later in the cardiac cycle than does the PA systolic wave, which is synchronous with the T wave on the electrocardiogram.

The sequence of the above steps will vary with the clinical context. The approach is also incomplete. For example, sudden distress may have no cardiopulmonary cause. Instead, it may originate in a blocked Foley catheter leading to an overdistended bladder.

Based on the initial assessment, a physician diagnoses the most likely cause of a change in a patient's condition. Table 53-1 lists the most common causes of sudden respiratory distress in a ventilated patient.

Table 53-1: Causes of Sudden Respiratory Distress in a Patient Receiving Mechanical Ventilation

<i>Patient-related causes</i>
Artificial airway problems
Movement of the endotracheal tube
Cuff herniation external
Compression of the endotracheal tube by the cuff
Cuff leak
Endotracheal tube kinking
Foreign body
Tracheoesophageal fistula
Innominate artery rupture
Malpositioning of the nasogastric tube
Build-up of secretions
Bronchospasm
Pneumothorax
Pulmonary edema
Pulmonary embolism
Acute hypoxemia
Blood in the endotracheal tube
Dynamic hyperinflation
Abnormal respiratory drive
Alteration in body posture
Drug-induced problems
Abdominal distension
Agitation
<i>Ventilator-related causes</i>
Ventilator malfunction
External ventilator circuit
Leaks or disconnects
Condensate
Inline nebulizers
Inadequate ventilator support
Patient-ventilator dyssynchrony

Artificial Airway Problems

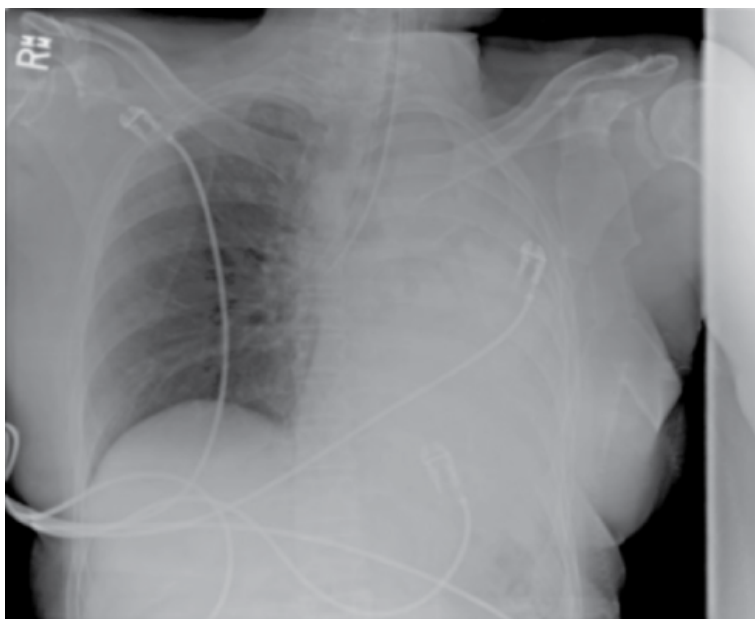
Several different problems with artificial airways can produce acute respiratory distress in the ventilated patient.

Movement of the Endotracheal Tube

An endotracheal or tracheostomy tube that is initially properly positioned may subsequently move up or down in a patient’s airway. Downward movement into a main-stem bronchus (endobronchial intubation) is estimated to occur in approximately 10% of ventilated patients.^{9–11} It is usually caused by inadequate external fixation of the endotracheal tube, excessive neck movement, or both.¹²

Main-stem intubation occurs more often on the right side, because the right main-stem bronchus forms a less-acute angle with the trachea than does the left main-stem bronchus¹¹ and because the bevel of most endotracheal tubes is directed rightward (Fig. 53-6). The lack of ventilation to the contralateral lung produces atelectasis, which, in turn, causes shunting of blood and hypoxemia. The delivery of a high volume to the intubated lung predisposes to the development of a pneumothorax.⁴

Figure 53-6



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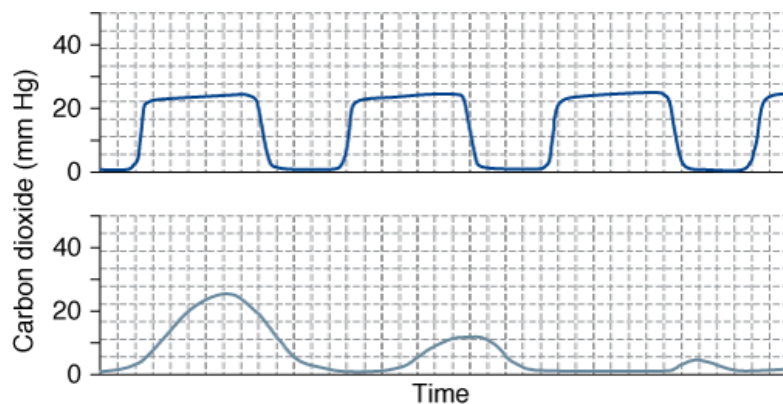
Right main-stem intubation. The endotracheal tube was inserted too far down, resulting in intubation of the right main-stem bronchus and atelectasis of the left lung with resultant mediastinal shift toward to the left.

Clues to the development of endobronchial intubation include an increase in peak airway pressure accompanied by a proportional increase in plateau pressure, asymmetric expansion of each hemithorax, and a decrease in breath sounds over the contralateral lung. In a prospective study, however, Brunel et al¹⁰ found that clinicians considered breath sounds equal over both lung fields in 60% of patients with an endobronchial intubation.

If endobronchial intubation is suspected, the endotracheal tube should be pulled back a few centimeters, the chest auscultated again, and the position of the tube confirmed by radiography. The problem can be prevented by securing the tip of the tube at least 3 to 4 cm above the carina, and obtaining a radiograph at the time of intubation.¹² In addition, centimeter markings on an endotracheal tube are helpful as a reference, although not completely reliable.¹⁰ In general, an endotracheal tube should be secured at reference markings of 23 cm in men and 21 cm in women.¹⁰ On the chest radiograph, the tip of the endotracheal tube should be located at a level corresponding to the top of the aortic knob when the head is in a neutral position.¹³

An endotracheal tube may also migrate above the vocal cords, or get dislodged into the esophagus.¹¹ Esophageal intubation may arise as a result of attempts to blindly reposition an endotracheal tube. Malpositioning causes sudden distress accompanied by phonation, audible escape of air through the nose and mouth, absence of tube condensation, decrease in tidal volume (V_T), decrease (or increase) in peak airway pressure, and abdominal distension. Chest radiography can be misleading because the trachea overlays the esophagus, making it difficult to tell whether the tube is located in the esophagus or trachea. A reading of zero on an end-tidal CO_2 monitor helps in detecting esophageal intubation (Fig. 53-7). Upward movement of an endotracheal tube usually results from excessive neck movement and/or inadequate tube fixation. Conrardy et al¹² found that neck extension caused an endotracheal tube to move an average of 1.9 cm away from the carina, but movement was as much as 5.2 cm in some patients. This observation may explain why patients who are carefully restrained can still self-extubate.

Figure 53-7



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End-tidal CO₂ tracing and esophageal intubation. *Upper panel.* End-tidal CO₂ tracing in a patient in whom an endotracheal tube was inserted correctly in the trachea. During inhalation, end-tidal CO₂ is 0 mm Hg. During exhalation, the CO₂ tracing rises steeply, attains a near plateau, and then quickly returns to the baseline. The end tidal CO₂ in this patient, 25 mm Hg, is below the normal range of 35 to 45 mm Hg. *Lower panel.* End-tidal CO₂ tracing in a patient in whom an endotracheal tube was erroneously inserted in the esophagus. The expected shape of the CO₂ tracing has been replaced by small transient rises in the CO₂ tracing, which progressively decrease with each squeeze of the Ambu bag.

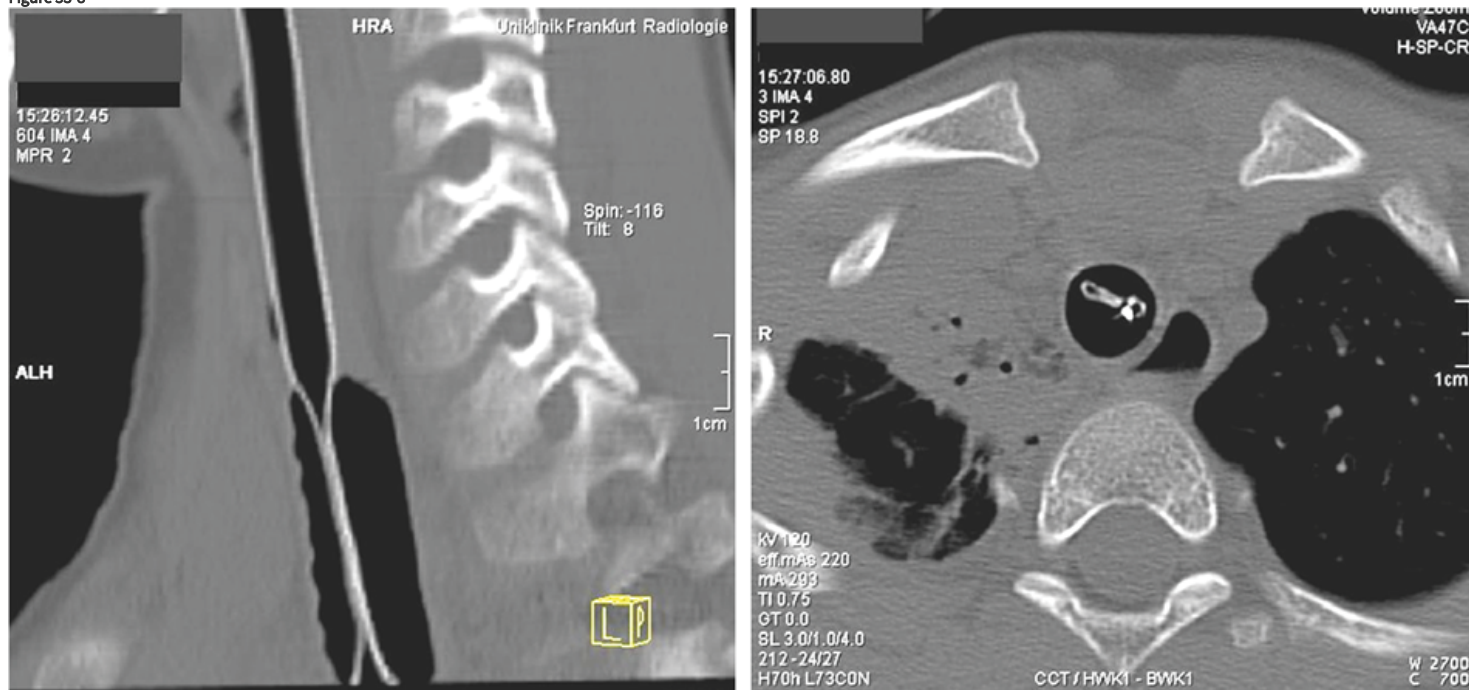
Cuff Herniation

Herniation of the cuff of an endotracheal tube over its distal tip can cause complete airway obstruction.¹¹ This problem most commonly occurs after changes in position of the tube or changes in the posture of the head and neck. Clues to cuff herniation include an increase in peak airway pressure, resistance during manual ventilation, a decrease in V_T, difficulty in passing a suction catheter, and an abnormal musical sound during inspiration.² Deflation of the cuff produces immediate relief. In a stable patient with an unclear source of airway obstruction, the diagnosis may not be made without bronchoscopy.

External Compression of the Endotracheal Tube by the Cuff

External compression of an endotracheal tube as a result of an inflated cuff is a rare cause of airway obstruction (Fig. 53-8).¹⁴

Figure 53-8



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External compression of endotracheal tube. Computed tomography images of the sagittal plane of the cervical spine and trachea (*left panel*) and transversal plane at the level of the first thoracic vertebra (*right panel*) in an 8-year-old boy who required endotracheal intubation as a result of multiple injuries received in a traffic accident. The inflated cuff produced critical obstruction of the endotracheal tube. The *white dot* on the scan in the right panel indicates the obstructed lumen of the tracheal tube. (Used, with permission, from Hofstetter et al.¹⁴)

Cuff Leak

The failure of the cuff on an endotracheal tube to form a seal with the wall of the trachea leads to gas leakage, which may or may not be of clinical consequence.¹¹ A large leak may cause the ventilator alarms to sound, signaling a low peak airway pressure and low exhaled V_T , and cause alveolar hypoventilation. Other clues include ability to phonate, frothy secretions in the patient's mouth, hearing a gurgle over the trachea or larynx on auscultation, a discrepancy between expired and inspired V_T , and failure to maintain a set level of PEEP.¹¹

The most common cause of cuff leak is malposition of the endotracheal tube. Consequently, the development of a cuff leak should cause the physician to reevaluate tube position.

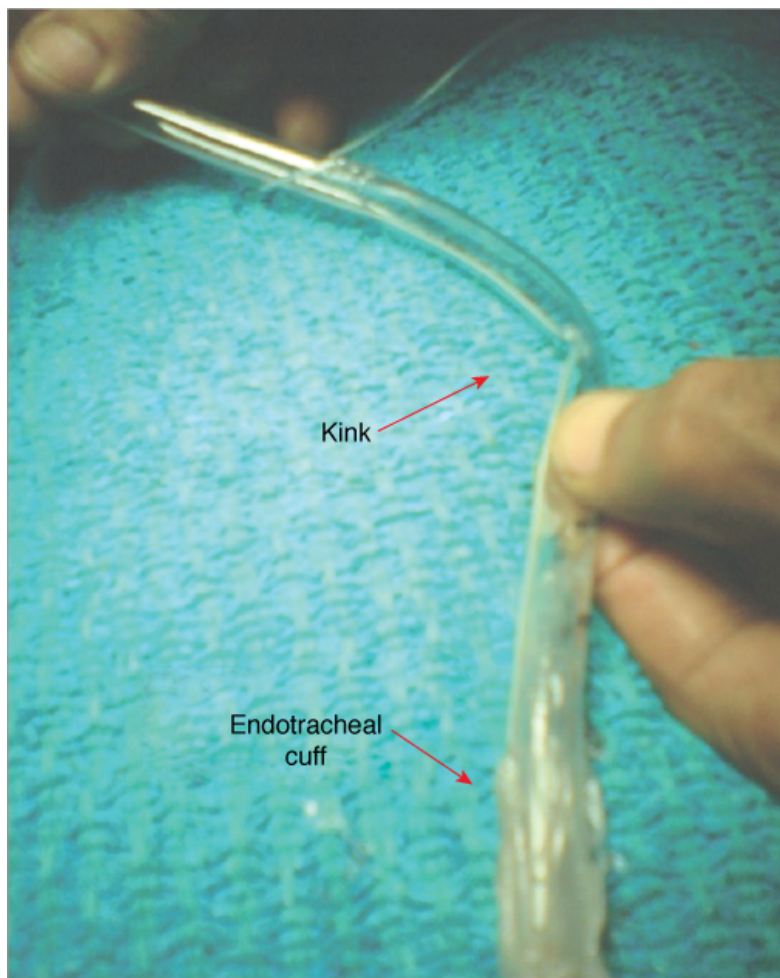
The cuff may be deflated not only because of a leak in the cuff itself, but also because of a leak in the pilot balloon or in the external valve assembly.¹³ A clamp placed on the cuff inflation tubing proximal to the valve and pilot balloon can temporarily correct this problem (without having to immediately replace the endotracheal tube).

Complete rupture of a cuff can lead to aspiration of saliva, vomit, or food; the entire endotracheal tube needs to be replaced.

Endotracheal Tube Kinking

The endotracheal tube can become kinked as a result of changes in the position of the head and neck.¹⁰ Some authors have recommended immersing an endotracheal tube in warm saline as a means of decreasing the incidence of mucosal trauma and epistaxis following nasal intubation,¹⁵ but this practice has been reported to give rise to subsequent kinking.¹⁶ Kinking may also occur if the tube softens through being warmed by heated humidified gas,¹³ or when the patient bites the tube. Figure 53-9 shows an instance of endotracheal-tube kinking that was attributed to earlier movement of the tube from one angle of the mouth to the other.¹⁷ We speculate that this shift in position of the tube might have led to a force acting in a direction perpendicular to the torque of the natural curve of an endotracheal tube, and thus induced kinking. Kinking leads to an increase in peak airway pressure and difficulty in passing a suction catheter. When this occurs, the endotracheal tube should be withdrawn and replaced.

Figure 53-9



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A kink found approximately 10 cm above the cuff of an endotracheal tube in a direction opposite to the natural curvature of the tube (toward the convex side) that resulted approximately an hour after the tube had been moved from the right angle of mouth and fixed on the left angle. (Used, with permission, from Hariharan et al.¹⁷)

Foreign Body

The endotracheal tube can become occluded with a variety of foreign bodies, including dried lubricant,¹⁸ surgical tape,¹⁸ a broken stylet sheath,¹⁹ plastic debris retained from the time of manufacturing the Murphy eye in the tube,²⁰ and nasal turbinates avulsed during nasotracheal intubation.²¹

Tracheoesophageal Fistula

Tracheoesophageal fistula is a rare complication that is primarily caused by ischemia of the tracheal wall secondary to pressure from an overinflated cuff or by movement of a tracheal tube or its tip.²² Patients with this complication typically have had both a tracheal tube and a nasogastric tube in place for a long time.^{11,23}

Clues to the presence of a tracheoesophageal fistula include the inability to deliver a preset V_T despite a functional cuff, gastric distension, audible leak of air through the mouth, copious airway secretions, the suctioning of gastric contents from the lower respiratory tract, and clinical deterioration and coughing on deflation of the cuff.¹³ Diagnosis needs to be verified by bronchoscopy with direct visualization of the esophagus or by cine-esophagography or cinetracheography.¹³ Most experts believe that closure of the fistula demands operative repair; opinions vary as to optimal timing.¹³

Innominate Artery Rupture

Erosion of the innominate artery has been reported to occur in 0.4% to 4.5% of tracheostomies.²⁴ Most ruptures occur at some point during the first three weeks after a tracheotomy,²⁴ with peak incidence 7 to 14 days after the procedure.²⁵ The underside of the tracheal tube, the tracheal cuff, or the tip of the tube erode through the anterolateral tracheal wall, at the point where the innominate artery crosses the trachea. Risk factors include excessive movement of the tube, excessive movement of the neck, overinflation of the cuff, or radiotherapy. Several authors emphasize that making a tracheotomy incision below the third tracheal ring increases the risk of fistula formation, but the complication can occur even when the incision is placed between the second and third tracheal rings, as recommended.²⁵

Any peristomal bleeding or hemoptysis in a patient with a recent tracheotomy should lead to detailed assessment to ascertain the underlying cause. Bleeding within 48 hours is usually associated with local factors such as traumatic puncture of anterior jugular or inferior thyroid veins, mucosal abrasion from a suction catheter, or systemic coagulopathy. Bleeding occurring 3 days to 6 weeks after a tracheotomy should be judged to be the result of innominate artery rupture until proven otherwise.²⁵ The clinical presentation may be quite dramatic, with blood gushing from the tracheal tube. It may be heralded by pulsation of the tracheal cannula or a “sentinel bleed,” which is reported to occur in more than 50% of patients.²⁵

Adequate oxygenation is the mainstay of immediate management with simultaneous identification and termination of bleeding.²⁵ The tracheostomy cuff should be overinflated in an attempt to achieve tamponade, and the cuffed tube advanced so that the balloon lies distal to the tracheostomy stoma. An attempt should be made to digitally compress the artery by inserting an index finger into the stoma and compress the artery against the posterior surface of the manubrium.²⁴ If this is successful, a blood transfusion should be administered while the patient is being transported to the operating room for sternotomy and ligation of the vessel. Time should not be wasted in performing imaging studies. Mortality is extremely high: greater than 85% in a review of the world literature.²⁴

Other Problems

Malposition of the Nasogastric Tube

A stiff nasogastric tube can easily bypass the cuff of an endotracheal tube and enter the airway. If the nasogastric tube is connected to suction, the continuous negative pressure will create a constant flow of gas out of the airways.²⁶ Consequently, the low V_T alarm will sound. The negative pressure can also cause triggering of the ventilator. In particular, ventilator triggering in a patient who is receiving a paralytic agent suggests that a nasogastric tube is misplaced in the airway. Other complications of nasogastric tubes include passage of the tube through the tracheobronchial tree into the pleural space, causing the formation of a bronchopleural fistula; infusion of nutritional solutions through the misplaced nasogastric tube can cause an empyema. A nasogastric tube can also cause esophageal perforation leading to pneumomediastinum, pneumothorax, and tracheoesophageal fistula.

Secretions

Airway secretions can cause problems by being too copious or too dry. Excessive secretions can lead to mucus plugging and atelectasis. To avoid this problem, careful bronchial toilet and frequent suctioning are necessary in patients with copious secretions. If atelectasis occurs and fails to resolve with conservative measures, bronchoscopy should be undertaken.

A helpful clue to the presence of excessive secretions is the presence of a sawtooth pattern on the flow tracing.⁷ In a study of fifty intubated patients, Jubran and Tobin⁷ found that the presence of a sawtooth pattern was approximately six to eight times more likely in patients who had secretions than in patients without secretions. Conversely, a smooth flow-volume curve was about one-quarter as likely to be found in patients with secretions as in patients without secretions. Clinical examination had much higher false-positive and false-negative rates (42% and 43%, respectively) than the flow-volume curves (12% and 14%, respectively). The usefulness of a sawtooth pattern for detecting secretions was confirmed by Guglielminotti et al²⁷ in a study of sixty-two patients who were receiving pressure-support or assist-control ventilation.

A tracheal tube bypasses the upper airway, which normally heats and humidifies inspired gas. Consequently, secretions may become excessively dry and encrusted. Inspissated secretions can cause significant blockage of the tracheal tube over a relatively short period.²³ Clues to this problem include an increase in peak airway pressure without an associated increase in plateau pressure, and difficulty in passing

a suction catheter. Obstruction appears to be a greater problem when heat and moisture exchangers are employed, as opposed to hot water humidifiers (see [Chapter 51](#)).

When secretions dry and accumulate, they can cause complete obstruction or ball-valve obstruction of the endotracheal tube. When this occurs at the distal tip of the tube, positive pressure during inspiration will cause the mass to move away from the tip, permitting the flow of gas. During expiration, expiratory pressure will push the mass of dried secretions into the distal port producing occlusion and cessation of airflow. The ball-valve obstruction causes pulmonary hyperinflation and may produce a tension pneumothorax.²⁸

Bronchospasm

Bronchoconstriction is a common cause of sudden respiratory distress. Clues to its development include an increase in peak airway pressure with little or no change in plateau pressure. Clinical manifestations include wheezes, pulsus paradoxus, and evidence of increased work of breathing (recession of the suprasternal space, supraclavicular fossae, and intercostal spaces; heightened activity of the accessory muscles; tracheal tug; and paradoxical motion of the rib cage and abdomen). Management includes bronchodilator therapy.

Pneumothorax

Sudden respiratory distress in a ventilator-supported patient should always arouse suspicion of a pneumothorax, because 60% to 90% of such pneumothoraces are reported to be under tension ([Fig. 53-10](#)).²⁹[Chapter 44](#) provides a full discussion of barotrauma and its pathogenesis.

Figure 53-10



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Development of a large left pneumothorax in a mechanically ventilated patient. Portable anteroposterior chest radiograph showing large (tensionlike) pneumothorax with significant displacement of heart and mediastinum to the opposite side. Vascular congestion of the right lung also noted.

A pneumothorax is typically associated with an increase in peak airway pressure and a proportional increase in plateau pressure. Other manifestations include hyperresonance, tracheal deviation to the contralateral side, decreased breath sounds, and cardiovascular collapse. If a pneumothorax is suspected and death is imminent, a 14- to 16-gauge needle should be inserted into a second intercostal space. If a gush of air is heard, a thoracostomy tube is inserted. If the patient is stable, however, a chest radiograph should first be performed to verify the diagnosis before inserting a chest tube.

Pulmonary Edema

Pulmonary edema can cause sudden distress associated with an increase in plateau pressure and a smaller increase in the gradient between the peak and plateau pressures. When present, pink, frothy secretions and an increase in pulmonary artery wedge pressure are particularly helpful.

Pulmonary Embolism

Acute pulmonary embolism is an uncommon cause of sudden respiratory distress in a ventilated patient. Typical manifestations include dyspnea, tachypnea, chest pain, fever, hemoptysis, pleural rub, and features of deep vein thrombosis. An important clue, when it occurs, is sudden hypoxemia with no change in peak airway pressure—a unique combination. The risks associated with transporting a critically ill patient alter the approach to diagnosis. Bedside duplex ultrasonography of the leg veins is a reasonable initial test.³⁰ Although ventilation–perfusion scans are difficult to perform in a ventilated patient, they have reasonable diagnostic reliability.³¹ Helical computed tomographic scanning is worthwhile, although doubts have been raised about its accuracy in this setting.³²

Acute Hypoxemia

Some ventilated patients present with acute hypoxemia, which may or may not be accompanied by acute distress. The differential diagnosis for new onset of hypoxemia can be considered in terms of ventilator-related problems, progression of the underlying disease, onset of a new medical problem, effects of interventions and procedures, and medications.³³ Table 53-2 lists the causes of worsening oxygenation in the ventilated patient.

Table 53-2: Causes of Worsening Oxygenation in the Ventilated Patient

<i>Ventilator-related problems</i>
Airway malfunction
External circuit malfunction
Ventilator malfunction
Inappropriate ventilator settings
<i>Progression of an underlying disease process</i>
Acute respiratory distress syndrome
Cardiogenic pulmonary edema
Pneumonia
Sepsis
Acute exacerbation of asthma or chronic obstructive pulmonary disease
<i>Onset of a new problem</i>
Pneumothorax
Atelectasis
Aspiration (gastric or oropharyngeal)
Ventilator-associated pneumonia
Sepsis
Pulmonary thromboembolism
Fluid overload
Bronchospasm
Retained secretions
Shock
Seizure
<i>Effects of interventions and procedures</i>
Endotracheal suctioning
Changes in body position
Chest physiotherapy
Bronchoscopy
Thoracentesis
Peritoneal dialysis
Hemodialysis
<i>Medications</i>
Bronchodilators
Vasodilators
β-blockers

Blood from the Endotracheal Tube

The return of bright red blood from a tracheal tube is alarming.³⁴ It is most commonly caused by trauma from a suction catheter, especially in patients with inflamed airways. Trauma secondary to the cuff also causes bleeding: The most dramatic form is innominate artery fistula. Airway bleeding may arise with infection, such as necrotizing pneumonia or tracheobronchitis. Pulmonary hemorrhage may be part of the underlying disease, such as Goodpasture syndrome, Wegener granulomatosis, neoplasm, or disseminated intravascular coagulation. Blood mixed with pink, frothy secretions suggests pulmonary edema. Pulmonary embolism accounts for a small proportion of patients with hemoptysis.

Pulmonary artery rupture can arise as a complication of pulmonary artery catheterization.³⁵ Rupture usually results from inflating the balloon when the catheter is too far distal. Risk factors include advancing the catheter with the balloon deflated, failure to observe the pulmonary artery pressure while the balloon is being inflated (inflation should cease immediately when a wedged tracing is obtained), hand

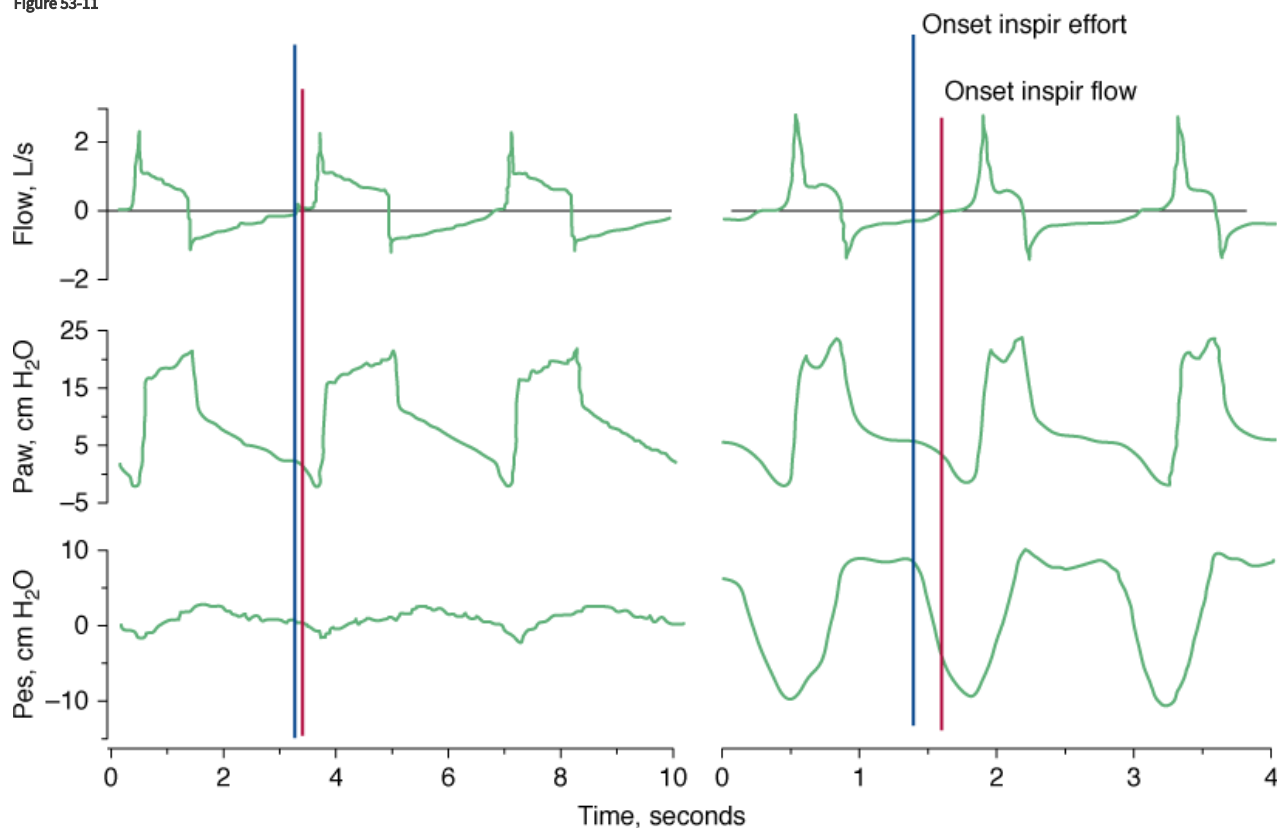
flushing of the catheter while it is wedged, and pulmonary hypertension. If pulmonary artery rupture is suspected, the patient should be placed in a lateral decubitus position with the affected lung down, a double-lumen endotracheal tube inserted (to protect the contralateral lung), and arrangements made for surgical intervention.

Dynamic Hyperinflation

Ventilator-supported patients commonly display tachypnea and an increased time constant (secondary to increased respiratory resistance and/or low pulmonary compliance), which predisposes to the development of dynamic hyperinflation.³⁶ This hyperinflation is associated with the presence of a positive recoil pressure at the end of expiration, termed *intrinsic PEEP* (*PEEPi*) or *auto-PEEP*.^{37,38}

Hyperinflation may cause significant patient discomfort for at least two reasons. One, it decreases the efficiency of force generation by the respiratory muscles. Two, it increases work of breathing because inspiration occurs at the upper, less compliant portion of the pressure–volume curve of the lung, where inwardly directed elastic recoil of the chest wall poses an additional elastic load.^{39,40} To initiate inspiratory gas flow, a patient has to generate a negative pressure equal in magnitude to the opposing elastic recoil pressure (level of *PEEPi*) (Fig. 53-11). Likewise, if a patient with *PEEPi* is triggering a ventilator, the patient has to generate a negative-pressure equal in magnitude to the level of *PEEPi* in addition to the set minimum circuit pressure drop before a ventilator-assisted breath is initiated. This is one factor that accounts for the common occurrence of failure to trigger despite obvious respiratory effort (Fig. 53-12).^{41,42}

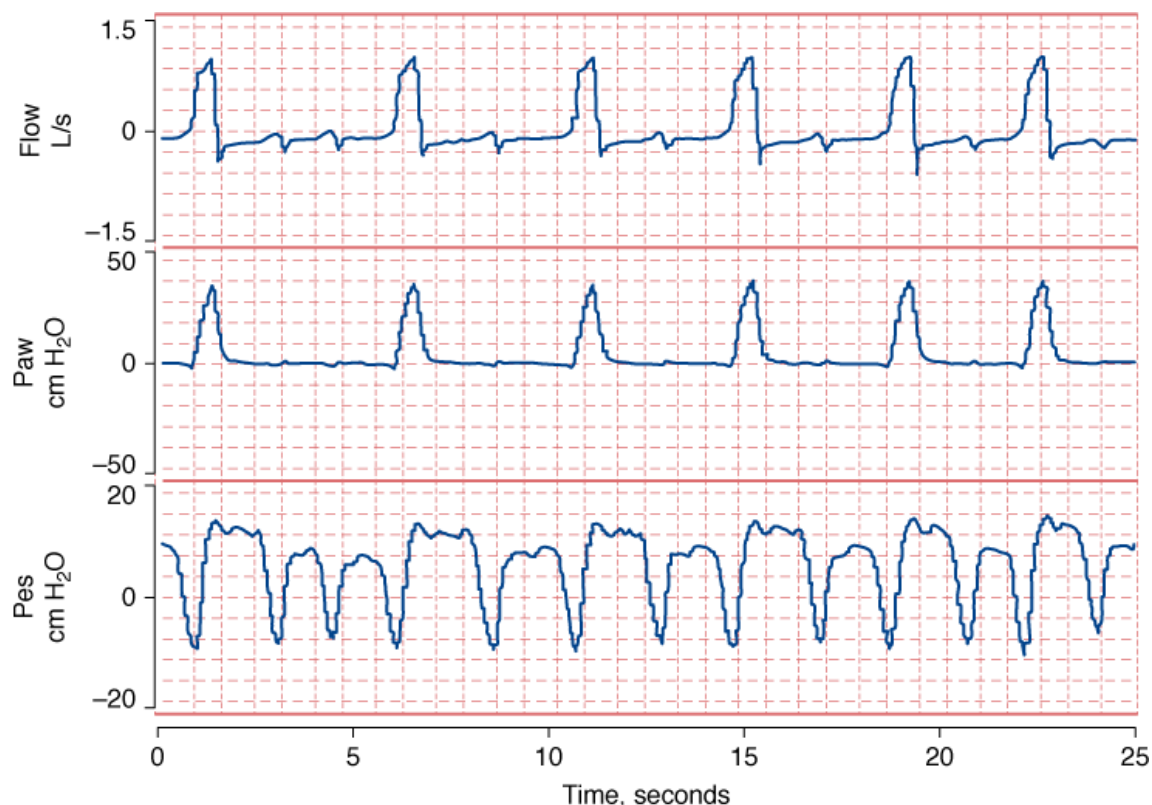
Figure 53-11



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Differing degrees of intrinsic positive end-expiratory pressure (*PEEPi*) in patients with chronic obstructive pulmonary disease. Tracings are flow, airway pressure (*Paw*), and esophageal pressure (*Pes*) in two patients with chronic obstructive pulmonary disease receiving assisted ventilation with pressure support set at 20 cm H₂O (no external PEEP is applied). The patient on the left had a myocardial infarction complicated by congestive heart failure, and the patient on the right had sepsis. *PEEPi*, estimated as the difference in esophageal pressure between the onset of inspiratory (*Inspir*) effort (vertical blue line) and the onset of inspiratory flow (vertical red line), was 0.5 cm H₂O in the patient on the left and 10.6 cm H₂O in the patient on the right. This method of estimating *PEEPi* is based on the assumption that the change in esophageal pressure reflects the inspiratory muscle pressure required to counterbalance the end-expiratory elastic recoil of the respiratory system.

Figure 53-12



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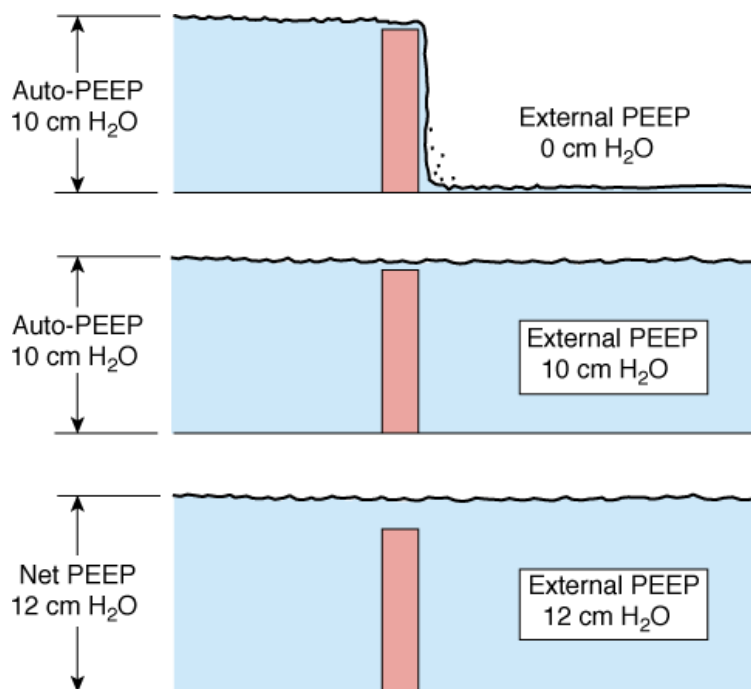
Failure to trigger the ventilator. Flow, airway pressure (P_{aw}), and esophageal pressure (P_{es}), in a patient with chronic obstructive pulmonary disease who is receiving assist-control ventilation at the following settings: tidal volume 600 mL, inspiratory flow 60 L/min, trigger sensitivity -2 cm H_2O , and positive end-expiratory pressure 0 cm H_2O . The patient's intrinsic respiratory rate is 28 breaths/min, whereas the number of breaths delivered by the ventilator is 16 breaths/min. That is, 43% of the patient's inspiratory efforts fail to trigger ventilator assistance. Contractions of the inspiratory muscles during the failed triggering attempts cause a temporary deceleration of expiratory flow and much less obvious decreases in airway pressure. The temporary decelerations in expiratory flow are followed by temporary accelerations of expiratory flow that coincide with the termination of the unsuccessful inspiratory effort.

Therapeutic measures to decrease PEEPi include bronchodilator agents, employment of a large-bore endotracheal tube; decreasing the minute ventilation by controlling fever or pain; and minimizing the ratio of inspiratory time to expiratory time by increasing inspiratory flow rate, or using nondistensible tubing in the ventilator circuit.

Addition of external PEEP can also decrease work secondary to the inspiratory threshold load effect of PEEPi.^{38,43} To initiate inspiratory flow, alveolar pressure must be decreased below ambient pressure. During normal spontaneous breathing, this is accomplished by only a small decrease in pleural pressure. For alveolar pressure to fall below ambient pressure in the presence of hyperinflation, however, a much greater decrease in pleural pressure is required. If ambient pressure is elevated by the application of external PEEP, inspiration is more easily accomplished because alveolar pressure needs to be decreased only below the level of external PEEP (rather than below zero). This may seem paradoxical: External PEEP, which is commonly used to induce hyperinflation in patients with microatelectasis, is being used to decrease the work of breathing induced by hyperinflation.

This paradox can be understood by considering the analogy of a waterfall over a dam.⁴⁰ The height of the waterfall represents the critical closing pressure of airways in patients with PEEPi and chronic obstructive pulmonary disease (COPD) (Fig. 53-13). Thus, elevating downstream pressure, as with external PEEP, has no influence on either expiratory airflow or the pressure upstream (PEEPi) from the site of critical closure (Fig. 53-13, upper panel). This situation exists until downstream pressure is elevated to a value equal to the critical closing pressure (Fig. 53-13, middle panel). Once downstream pressure is elevated above the critical closing pressure (the height of the waterfall), however, the pressure upstream increases immediately, and hyperinflation is exacerbated (Fig. 53-13, lower panel).

Figure 53-13



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The analogy of a waterfall over a dam (indicated by the *solid block*) is used to explain the effect of external PEEP (downstream pressure) on PEEPi (upstream pressure) during expiration. Elevation of downstream pressure with external PEEP has no influence on either expiratory flow or upstream pressure (PEEPi) until it is equal to the critical closing pressure. When downstream pressure exceeds the critical closing pressure, the pressure upstream increases and hyperinflation is exacerbated. (Used, with permission, from Tobin and Lodato.⁴⁰)

The counteracting effect of external PEEP operates only in the setting of airflow limitation. In the absence of airflow limitation, or when PEEPi results from expiratory muscle activity, the addition of external PEEP is a hindrance and adds to work of expiration. It is important to keep in mind that while external PEEP can help decrease inspiratory work, it does nothing to relieve the accompanying hyperinflation, which has other detrimental effects.³⁹ Another caveat is the heterogeneous distribution of lung units in patients with airflow limitation, each with its own critical closing pressure.⁴⁴ Given the regional inhomogeneities among lung units, the average level of PEEPi must exceed the lowest regional alveolar pressure. Thus, external PEEP equal to PEEPi could cause hyperinflation in the faster-emptying lung units.^{45–47} To minimize this risk, it is prudent to limit the level of external PEEP to approximately 70% of the level of PEEPi.

The importance of PEEPi as a cause of electromechanical dissociation during cardiopulmonary resuscitation is now well recognized.^{48,49} Rosengarten et al⁴⁸ reported a dramatic example. A young woman with acute severe asthma was intubated 25 minutes after initial presentation. Ventilation was commenced with a V_T of 8 to 10 mL/kg and a respiratory rate of 10 to 14 breaths/min. Oxygen saturation increased momentarily. Five minutes after intubation, blood pressure could not be recorded. External cardiac massage was instituted, but the patient remained in electromechanical dissociation. Twenty-five minutes after intubation, the situation was considered irretrievable. All resuscitation was stopped. Three minutes later, the patient returned to sinus rhythm, and the pulses became palpable. Ventilation was recommenced at 15 breaths/min. Within 15 seconds, the pulse disappeared. Ventilation was reduced to 6 to 8 breaths/min, and the pulse returned. Ventilation was transiently increased on three more occasions, and blood pressure disappeared each time. Seventy-five minutes after commencing resuscitation, the patient's blood pressure was 165/80 mm Hg. The patient suffered initial ischemic brain damage, but eventually made a close to complete recovery. In retrospect, it is clear that this patient suffered repeated episodes of circulatory arrest as a result of unrecognized ventilator-induced hyperinflation.

If faced with persistent hypotension in a patient with airflow limitation who is undergoing cardiopulmonary resuscitation, elevated PEEPi should be suspected. In such instances, the patient should be subjected to an apnea trial (after first ventilating the patient with 100% oxygen).¹³ Ventilation is stopped for 1 minute, and blood pressure measured. If blood pressure improves, ventilation is resumed, but aiming to achieve a lower minute ventilation.

Inadequate Respiratory Motor Output

Decreased respiratory neuromuscular output may cause sudden deterioration in a patient's respiratory status. Decreased respiratory motor output, however, is not typically accompanied by respiratory distress. Respiratory motor output may be decreased as a result of acute neurologic disorders, heavy sedation, or neuromuscular blocking agents.⁵⁰ A normal, or even elevated, motor output may be insufficient to maintain alveolar ventilation if ventilatory demands or the mechanical load suddenly increase.

Elevated Respiratory Motor Output

Excessive respiratory motor output may cause severe respiratory alkalosis, which, in turn, may cause arrhythmias, hypotension, cerebral vasoconstriction, and seizures.^{51,52} Ventilation can be stimulated by many physiologic, psychologic, and pathologic factors.⁵⁰ Extreme anxiety and panic cause dyspnea, fear, tachycardia, and palpitations, and when a patient misinterprets the basis of these symptoms, hyperventilation commonly develops.^{53,54} A patient's likelihood of hyperventilating in response to stress probably depends on biologic vulnerability, personality, and cognitive variables.⁵⁴

Several respiratory and nonrespiratory organic disorders stimulate breathing. The best known example is asthma, and even mild or moderate airway obstruction may be accompanied by a fall in partial pressure of arterial carbon dioxide (PaCO_2) to below 25 mm Hg.⁵⁵ The mechanism of the hyperventilation is uncertain: It is probably related to hyperinflation⁵⁶ and stimulation of vagal afferents,⁵⁷ and it is not reversed by bronchodilation.⁵⁸ The reduction in PaCO_2 is quantitatively consistent with hypoxic stimulation of the peripheral chemoreceptors, although supplemental oxygen does not restore ventilation to normal.⁵⁹ Respiratory stimulation also occurs in patients with COPD,⁶⁰ pulmonary fibrosis,^{60,61} pneumonia,⁶² pulmonary embolism,⁶³ pulmonary hypertension,^{60,64} and heart failure.⁶⁵ In asthma, the most profound hyperventilation occurs with relatively mild disease,^{55,66} and the same may apply with the other causes of respiratory stimulation. Pain stimulates breathing,⁶⁷ whereas hyperventilation increases the pain threshold.⁶⁸ Hyperventilation can be severe with certain drug overdoses, such as aspirin,⁶⁹ and it is a striking feature of sepsis⁷⁰ and major organ system failure,⁷¹ although it does not occur in thyrotoxicosis. Acute diabetic ketoacidosis stimulates the peripheral chemoreceptors to produce deep rapid breathing and hypocapnia.⁷²

Plum et al⁷³ coined the term *central neurogenic hyperventilation* to describe nine patients with tachypnea, of metronomic regularity, which persisted for hours or days at a time. Plum et al⁷⁴ subsequently revised their opinion, and attributed the heightened ventilation to stimulation of afferent peripheral reflexes arising in the lung and chest wall. It has not been possible to reproduce central neurogenic hyperventilation in animals. In their own clinical practice, Plum et al⁷⁴ are convinced of having seen only three patients with the condition, although brain histology was obtained in none of these cases. A few patients have been reported who displayed profound respiratory alkalosis, persisting during sleep or after administration of morphine, with normal oxygenation, normal chest radiography, and who had histologic evidence of pathology in the pons or some other area of the brain.^{75–80} The rarity of this disorder is supported by the data of North et al.⁸¹ Of 227 patients admitted to a neurosurgical unit, fifty-seven had a respiratory rate greater than 25 breaths/min and nine of these had a PaCO_2 less than 30 mm Hg. In all but one patient, however, the tachypnea and hypocapnia could be explained by hypoxemia associated with a pulmonary abnormality or metabolic acidosis. While hyperventilation in critically ill patients with an acute neurologic disorder is rarely neurogenic in nature, it nevertheless carries a poor prognosis.^{81–84}

Body Posture

An alteration in body posture can cause a fall in partial pressure of arterial oxygen (PaO_2) of as much as 30%,^{33,85} especially in patients with unilateral lung disease. The hypoxemia is secondary to the effect of gravity on blood flow, as blood is shunted through the diseased dependent lung.

Drug-Induced Deterioration

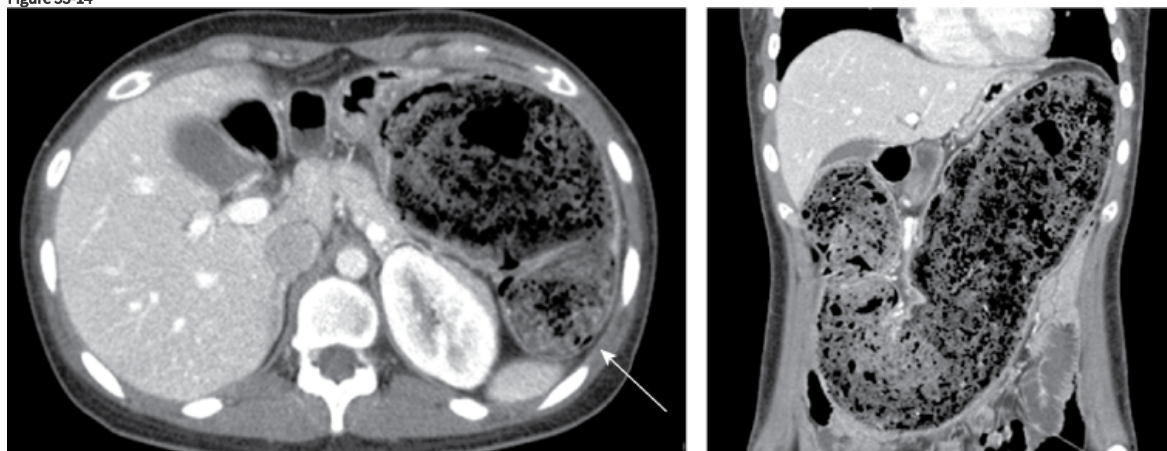
Hypoxemia may result from worsening of ventilation-perfusion relationships secondary to bronchodilators or vasodilators (e.g., nitroglycerin or nitroprusside). Administration of intravenous lipid compounds can also produce hypoxemia.³³ Aminoglycoside antibiotics can provoke or aggravate neuromuscular blockade and produce respiratory embarrassment.⁸⁶ High levels of theophylline may produce agitation and/or seizures.

Abdominal Distension

Abdominal distension can cause elevation of the diaphragm, basilar atelectasis, and deterioration in ventilation–perfusion relationships. Abdominal distension may result from gastric distension, ascites, peritoneal dialysis, or bowel perforation. Gastric distension may result from a prolonged or difficult attempt at intubation; elevation of mouth pressure above the lower esophageal sphincter pressure during the delivery of manual ventilation; or tracheal pressure exceeding the cuff pressure and lower esophageal sphincter pressure and the mouth remaining closed during mechanical ventilation.^{87,88} Massive distension, “meteorism,” can produce gastric rupture. Insertion of a small-bore nasogastric tube helps prevent or alleviate this complication.

Massive colonic distension without small intestinal dilation or distal obstruction (Ogilvie syndrome) has been described in patients receiving mechanical ventilation (Fig. 53-14).^{89,90} The mechanism is not known, but exaggerated aerophagia may be an important factor. Perforation of an overdistended cecum is often the first evidence of this complication. Accordingly, abdominal radiographs should be obtained in patients with abdominal distension, and cecostomy should be considered when the cecal diameter exceeds 9 cm. Massive colonic distension with or without massive ascites can also complicate *Clostridium difficile* colitis.⁹¹

Figure 53-14



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Ogilvie syndrome in a 31-year-old woman. Axial (left panel) and coronal (right panel) contrast-enhanced computed tomography images demonstrate a markedly distended transverse colon with large amount of fecal material. The arrow in the left panel points at the transitional zone (splenic flexure) that is devoid of any obstructive lesions. (Used, with permission, from Choi et al.⁸⁹)

Agitation

Ventilated patients develop agitation as a result of four major factors: pain, anxiety, delirium, and dyspnea. Innumerable factors can cause pain in a critically ill patient, ranging from simple factors such as poorly applied adhesive tape to the pain of a major surgical procedure. Feelings of anxiety, insecurity, fear, and panic are very common in ventilator-dependent patients,^{92,93} including a fear of impending death. Untreated pain and anxiety can lead to insomnia and delirium. These problems are aggravated by the discomfort of dyspnea, which may result from injudicious ventilator settings (see *Patient–Ventilator Dyssynchrony*). It is important to follow a systematic approach when evaluating a patient for the source of pain or anxiety, because simple problems, such as an overdistended bladder, may be overlooked while a search is conducted for a more esoteric source of distress.

Ventilator Malfunction

Resolution of a patient’s distress with the onset of bagging indicates that the problem lies within the ventilator or its external circuit (see Fig. 53-1). Malfunction of the ventilator may arise from tubing connected to a wrong outlet; a poor fit of connections; uncoupling of connections; defective material; obstruction of the circuit secondary to kinks, intraluminal fluids, or a malfunctioning valve; or malfunction of microprocessor controls. If malfunction is suspected, the ventilator should be replaced while each component is checked against the schematic circuit diagram. When a patient’s delivered V_T is adequate and distress is relieved by manual ventilation, a fault in the setting of the fractional inspired oxygen concentration should be suspected. This problem can be verified by obtaining an independent direct measurement of inspired oxygen concentration.

External Ventilator Circuit

The ventilator circuit primarily consists of the tubing and connectors that form inspiratory and exhalation limbs; the humidifier; the exhalation valve assembly and flow measuring devices; adaptors placed within the circuit for monitoring or delivering medications; and inline filters.¹³ Problems within this circuit that can cause patient distress include leaks, disconnects, accumulation of condensate, and inline nebulizers.

Leaks or Disconnects

A leak may arise in parts of the circuit assembly that screw together. Leaks or uncoupling of connections will cause sounding of the low pressure and low V_T alarms. The Y-connection to the tracheal tube is the most common site of a disconnection. Other locations of disconnections and leaks include the humidifier system, an incompetent exhalation valve assembly, disconnection of the proximal line that connects the pressure tap at the Y-piece to the ventilator manometer, or small ruptures in the tubing.

In patients ventilated with pressure support, a leak at any location in the circuit, including at the cuff of the endotracheal tube, predisposes to a unique problem.⁹⁴ During pressure support, the ventilator strives to maintain a preset level of pressure throughout inspiration. A leak, however, will tend to cause the airway pressure to fall. To prevent the fall in airway pressure, the ventilator increases inspiratory flow. The algorithm employed by many ventilators for terminating the time of lung inflation is a fall in inspiratory flow to an absolute value of 5 L/min (or a fall in flow of 75% from the peak value). The increase in flow being delivered by the ventilator means that inspiratory flow never falls to the threshold required for termination of inflation. This phenomenon results in the unremitting application of positive pressure, which is relieved by correcting the leak.

Condensate

Gas supplied to the inspiratory limb is typically warmed to 32°C (89.6°F) to 34°C (93.2°F) and fully saturated with water vapor. The gas cools as it passes through the inspiratory tubing, causing condensate to form. The condensate accumulates in a U-loop of the circuit. If enough condensate accumulates, movement of the water can cause the ventilator to trigger. During assist-control ventilation, excessive triggering can lead to barotrauma or hemodynamic compromise. During pressure-support or pressure-controlled ventilation, the resistance caused by the condensate may cause a decrease in achieved V_T for a set airway pressure.

Inline Nebulizers

The insertion of a continuous-flow nebulizer between the patient and the pressure sensor within the ventilator can lead to hypoventilation during pressure support or intermittent mandatory ventilation (IMV).⁹⁵ When a patient's mean inspiratory flow rate is less than the flow of gas used to power the nebulizer (6 to 10 L/min), airway pressure will not fall sufficiently to trigger the ventilator. Moreover, the continuous flow from the nebulizer creates a bias flow, which the ventilator interprets as forming part of the patient's minute ventilation. Consequently, the low minute volume alarm will not sound.

Patient–Ventilator Dyssynchrony

Patient–ventilator dyssynchrony can lead to considerable patient distress, and it also impedes the effectiveness of the ventilator in decreasing respiratory work. For the most effective unloading of the inspiratory muscles, the ventilator should cycle in synchrony with the patient's central respiratory rhythm. For perfect synchronization, the period of mechanical inflation must match the period of neural inspiratory time (the duration of inspiratory effort), and the period of mechanical inactivity must match the neural expiratory time.^{96,97} The interplay between the ventilator and the respiratory neuromuscular apparatus is complex, and problems can arise at several points in the respiratory cycle: the onset of ventilator triggering, the remainder of inspiration after triggering, the switch from inspiration to expiration, and the end of expiration.⁴²

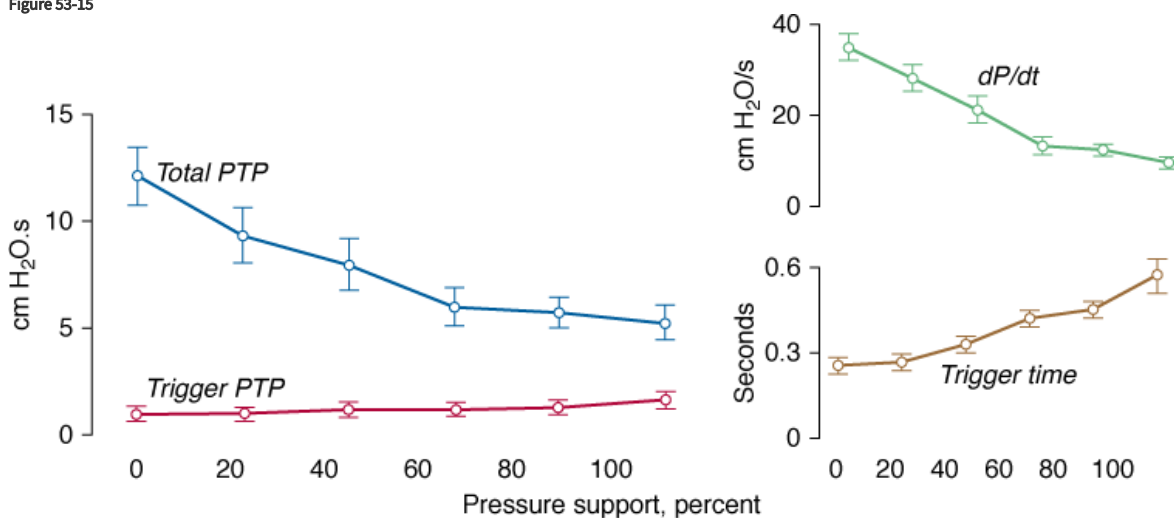
Triggering of the Ventilator

Patients reach the set sensitivity by activating their inspiratory muscles. For a given set sensitivity, the delay between onset of patient inspiratory effort and onset of ventilator assistance is a function of a patient's respiratory motor output. When a patient's respiratory drive is low, assistance may not begin until well into the patient's inspiratory time, thereby causing the ventilator to cycle almost completely out of

phase with the patient's respiratory cycle. When the threshold to open the demand valve is reached and the ventilator starts to provide positive-pressure assistance, the inspiratory neurons do not simply switch off, and a patient may expend considerable inspiratory effort throughout the machine-cycled inflation.⁹⁸ The level of patient effort during this post-trigger phase is closely related to a patient's respiratory motor output at the point of triggering.⁹⁹ As such, measures that decrease respiratory drive may enhance respiratory muscle rest during mechanical ventilation.

If respiratory motor output at the point of triggering is important, one might expect that effort during the time of triggering would determine patient effort during the remainder of inspiration.¹⁰⁰ To investigate this issue, Leung et al⁹⁹ applied graded levels of pressure support in eleven critically ill patients. They achieved a fourfold reduction in overall patient effort. Yet patient effort during the time of triggering did not change. The constancy of effort during the trigger phase was probably secondary to different factors becoming operational as the level of ventilator assistance was varied (Fig. 53-15). Thus, increases in the level of ventilator assistance do not substantially decrease patient effort during the time of triggering.

Figure 53-15



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Graded increases in pressure support produced a decrease in total pressure-time product (PTP) per breath (*blue symbols*), although PTP during the trigger phase (*red symbols*) did not change (*left panel*). The constancy of PTP during triggering probably resulted from different factors becoming operational at different levels of assistance (*right panel*). At low levels of pressure support, respiratory motor output (dP/dt *green symbols*) and intrinsic positive end-expiratory pressure (PEEPi) were high but triggering time was short (*brown symbols*), resulting in a large change in pleural pressure over a brief interval. At high levels of pressure support, dP/dt and PEEPi were low but triggering time was long, resulting in a smaller change in pleural pressure over a longer time. (Based on data from Leung et al.⁹¹)

Ineffective Triggering

At high levels of mechanical assistance, up to one-third of a patient's inspiratory efforts may fail to trigger the machine (see Fig. 53-12).^{99,101,102} The number of ineffective triggering attempts increases in direct proportion to the level of ventilator assistance.⁹⁹ Surprisingly, unsuccessful triggering is not the result of poor inspiratory effort. In a study of factors contributing to ineffective triggering, a decrease in the magnitude of inspiratory effort at a given level of assistance was not the cause; indeed, effort was 38% higher during nontriggering attempts than during the triggering phase of attempts that successfully opened the ventilator valve.⁹⁹ Significant differences, however, were noted in the characteristics of the breaths before the triggering and nontriggering attempts. Breaths before nontriggering attempts had a higher V_T than did the breaths before triggering attempts, 486 ± 19 and 444 ± 16 mL, respectively, and a shorter expiratory time, 1.02 ± 0.04 and 1.24 ± 0.03 seconds, respectively. An abbreviated expiratory time does not allow the lung to return to its relaxation volume, leading to an increase in elastic recoil pressure. Indeed, PEEPi was higher at the onset of nontriggering attempts than at the onset of triggering attempts: 4.22 ± 0.26 versus 3.25 ± 0.23 cm H₂O. Thus, nontriggering results from premature inspiratory efforts that are not sufficient to overcome the increased elastic recoil associated with dynamic hyperinflation.⁹⁹

When triggering fails as a result of dynamic hyperinflation, the nontriggering on that breath allows the lungs to more completely empty in preparation for the next breath. Occasionally, two or three failed triggering attempts take place before triggering becomes successful. Because there is no lung inflation during a failed triggering attempt, mechanical exhalation continues for a longer time and end-expiratory volume continues to fall until triggering becomes successful. This pattern may occur repeatedly such that it resembles the Wenckebach pattern of atrioventricular block on an ECG.¹⁰³

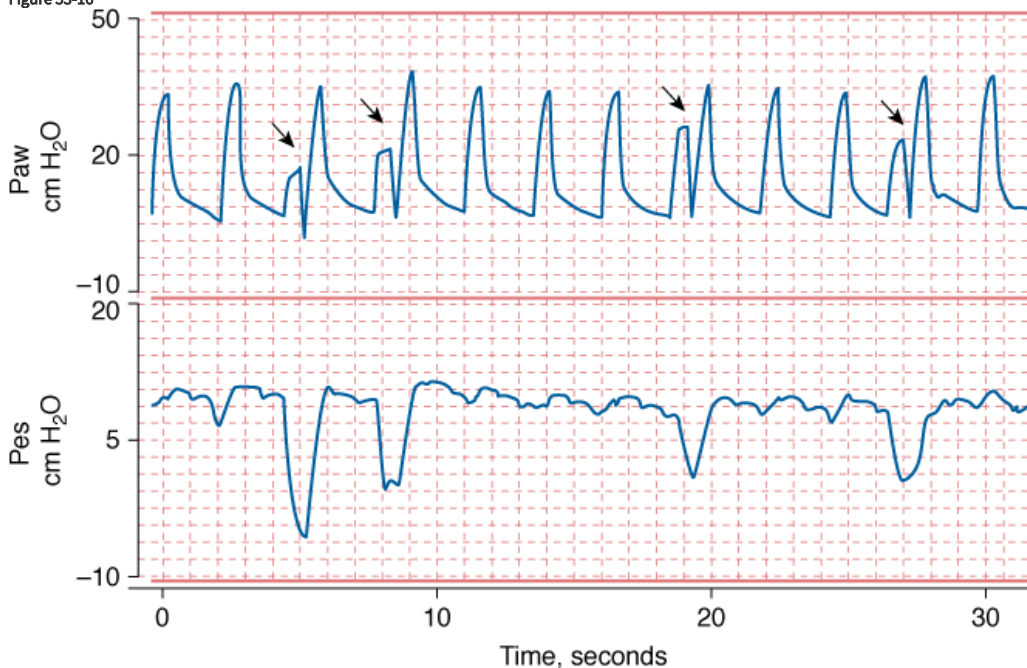
In addition to an increase in elastic recoil pressure,⁹⁹ an elevated PEEPi can also result from an increase in expiratory muscle activity. Parthasarathy et al⁹⁷ investigated the relative contributions of these two factors to ineffective triggering in healthy subjects receiving pressure support and in whom they induced airflow limitation with a Starling resistor. Nontriggering was linked to the fraction of PEEPi caused by elastic recoil but not to the fraction caused by expiratory effort. This observation suggests that external PEEP might be clinically useful in reducing ineffective triggering.

Although the magnitude of expiratory effort does not appear to influence the success of triggering attempts, the time that expiratory efforts commence in relation to the cycling of the ventilator is an important factor. Parthasarathy et al⁹⁷ quantified the relationship between the onset of expiratory muscle activity, measured with a wire electrode in the subject's transversus abdominis, and the termination of mechanical inflation by the ventilator. At pressure support of 20 cm H₂O, mechanical inflation was found to continue for a longer time into neural expiration in the breaths preceding nontriggering attempts. Continuation of mechanical inflation into neural expiration counters expiratory flow, and also decreases the time available for unopposed exhalation. Consequently, elastic recoil increases. In turn, a greater inspiratory effort will be needed to achieve effective triggering. In this way, the time that a patient commences an expiratory effort (in relation to cycling-off of mechanical inflation) partly determines the success of the ensuing inspiratory effort in triggering the ventilator.

Double Triggering

Some patients exhibit two mechanical inflations within a single neural inspiration, a phenomenon known as *double triggering* (Fig. 53-16). With assist-control ventilation, double triggering is likely when the set mechanical inspiratory time is substantially less than a patient's neural inspiratory time. In this situation, mechanical inflation terminates while the patient is still making an inspiratory effort. After a brief period, the ventilator may trigger again, resulting in a second inflation within the same neural inspiration and, thus, greater alveolar distension than with the delivery of a single tidal volume.¹⁰³

Figure 53-16



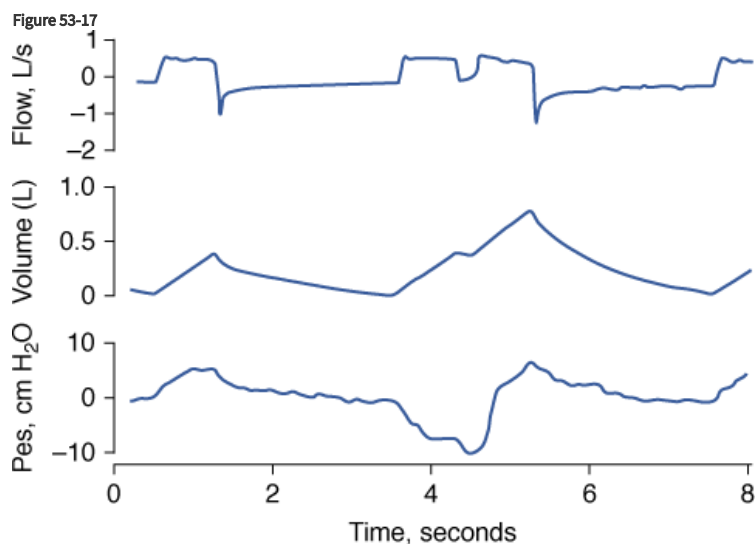
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Four incidents of double triggering, each indicated by an arrowhead. Airway pressure (P_{aw}) and esophageal pressure (P_{es}) in a patient with COPD and pneumonia who was receiving assist-control ventilation at the following settings: tidal volume 600 mL, inspiratory flow 60 L/min,

trigger sensitivity -2 cm H₂O, and positive end-expiratory pressure 5 cm H₂O. The duration of neural inhalation of the double-triggered breaths, roughly equivalent to the width of the associated swings in esophageal pressure, was substantially longer than the neural inhalation of the normally triggered breaths.

With pressure-support ventilation, double triggering is likely when the time constant is short (low resistance, high elastance) and patient neural inspiratory time is relatively long (a slow, spontaneous respiratory rate).¹⁰⁴ Following the loss of ventilator pressure after a first triggering attempt, volume decreases because the pressure exerted by the inspiratory muscles (P_{mus}) alone is not sufficient to sustain elastic recoil. During this phase, the persistence of neural inhalation will cause a progressive increase in P_{mus} , while elastic recoil continues to decrease. If the patient's neural inspiratory time is sufficiently long, a point is reached where P_{mus} will exceed elastic recoil and flow will become inspiratory and double triggering will occur.¹⁰⁵

Randomized clinical trials have revealed that use of a high tidal volume (12 mL/kg) is associated with an increased mortality in patients with the acute respiratory distress syndrome.¹⁰⁶ Consequently, it has become standard practice to lower the delivered tidal volume in order to minimize alveolar overdistension; for example, to keep the inspiratory pressure during a pause at the end of inspiration (plateau pressure) to 32 cm H₂O or lower.¹⁰⁶ A low tidal volume is typically accompanied by a short mechanical inspiratory time and thus these patients are especially susceptible to double triggering. Accordingly, conscious attempts to lower tidal volume can paradoxically result in greater alveolar distension than occurs with conventional tidal volume settings (Fig. 53-17). An additional (and often unrecognized) factor that may produce alveolar distension is the tachypnea-associated increase in intrinsic PEEP that accompanies lowering of tidal volume.¹⁰⁷



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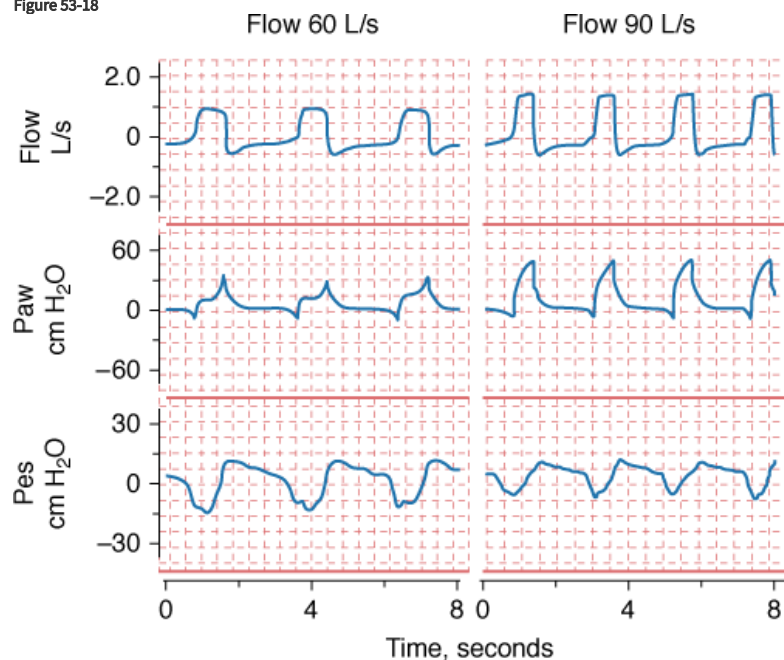
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Volume stacking caused by double triggering. Flow (*top panel*), volume (*middle panel*), and esophageal pressure (*Pes*; *lower panel*) in a patient with COPD receiving assist-control ventilation. During first breath, esophageal pressure remains positive indicating that the patient did not trigger the inflation. During the second breath, esophageal pressure becomes negative indicating active inspiratory effort, which lasts more than 1 second; the duration of mechanical inflation is 0.6 second. The longer duration of neural inspiration as compared with mechanical inflation causes the ventilator to deliver a second breath before there is time for exhalation. As a result, end-inspiratory lung volume increases (breath stacking) with a consequent increase in elastic recoil. The increase in elastic recoil is responsible for the higher peak expiratory flow on the second breath as compared with the first breath.

Setting of Inspiratory Flow

When a patient is first connected to a ventilator, inspiratory flow is set at some default value, such as 60 L/min. Many critically ill patients, however, have an elevated respiratory motor output and the initial flow setting may be insufficient to meet flow demands. As a result, patients will struggle against their own respiratory impedance and that of the ventilator (Fig. 53-18). Consequently, the work of breathing increases. Clinicians sometimes increase flow so as to shorten the inspiratory time and increase the expiratory time. But an increase in flow causes immediate and persistent tachypnea; as a result, expiratory time may be shortened.¹⁰⁸ In healthy subjects, Laghi et al¹⁰⁹ found that increases in inspiratory flow from 30 L/min to 60 and 90 L/min caused increases in the respiratory rate of 20% and 41%, respectively.

Figure 53-18



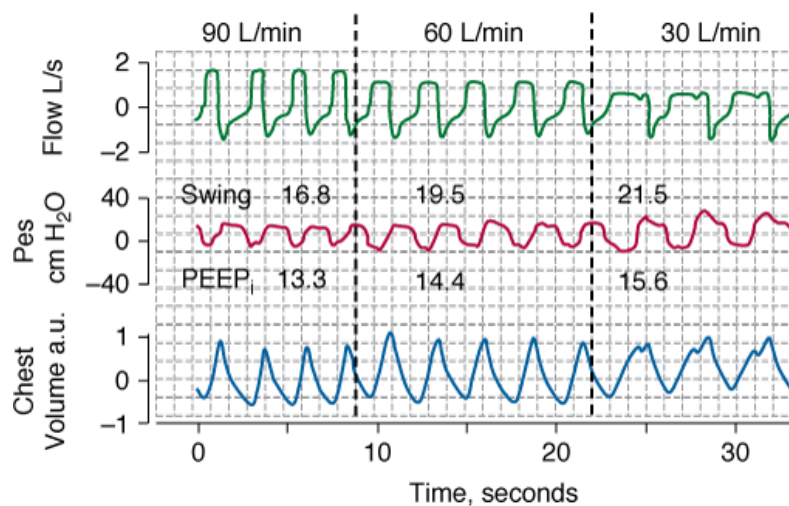
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Influence of ventilator flow setting on patient effort. Flow (inspiration directed upward), airway pressure (P_{aw}), and esophageal pressure (P_{es}) in a patient with respiratory failure who is receiving assist-control ventilation; inspiratory flow is set at 60 L/min in the *left panel* and at 90 L/min in the *right panel*. At an inspiratory flow of 60 L/min (*left panel*), the pronounced negative deflection in airway pressure (patient effort to trigger the ventilator) together with subsequent extensive scalloping signifies that the inspiratory flow delivered by the ventilator is insufficient to meet the high demand. At an inspiratory flow of 90 L/min (*right panel*), the small negative deflection in airway pressure together with the subsequent smooth convex contour signifies that the delivered flow satisfies the patient's respiratory drive. Accordingly, the flow of 90 L/min achieved greater unloading of the respiratory muscles, as signaled by the shorter duration of inspiratory effort and the smaller swings in esophageal pressure.

A main reason that clinicians increase inspiratory flow is to decrease inspiratory time, in the hope of allowing more time for expiration and thus decrease PEEP_i, especially in patients with COPD. Because increased flow usually leads to an increase in rate, the expected shortening of expiratory time might actually increase PEEP_i. Laghi et al¹¹⁰ studied this phenomenon in ten patients with COPD (Fig. 53-19). As with healthy subjects, an increase in flow from 30 to 90 L/min caused the respiratory rate to increase from 16.1 ± 1.0 to 20.8 ± 1.5 breaths/min. Despite the increase in rate, PEEP_i fell from 7.0 ± 1.3 to 6.4 ± 1.1 cm H₂O. The decrease in PEEP_i arose because of an increase in expiratory time, 2.1 ± 0.2 to 2.3 ± 0.2 seconds, which allowed more time for lung deflation. Why did expiratory time increase? An increase in inspiratory flow is usually achieved by shortening mechanical inspiratory time. The shortened inspiratory time combined with time-constant in homogeneity of COPD will cause overinflation of some lung units to persist into neural expiration. Continued inflation during neural expiration causes stimulation of the vagus nerve, which prolongs expiratory time.^{111,112}

Figure 53-19



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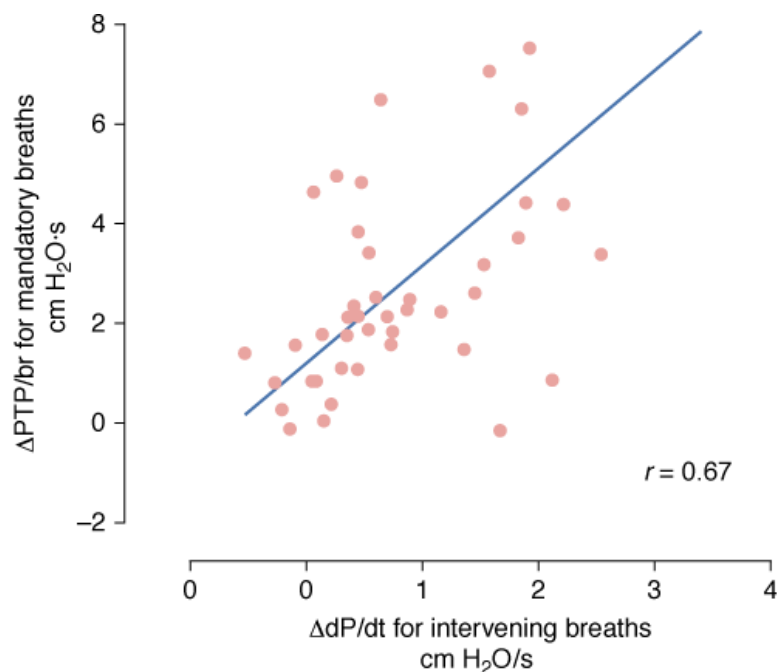
Continuous recordings of flow, esophageal pressure (P_{es}), and the sum of rib cage and abdominal motion in a patient with COPD receiving assist-control ventilation at a constant tidal volume. As flow increased from 30 to 60 and 90 L/min (from right to left; i.e., the opposite of the usual presentation), frequency increased (from 18 to 23 and 26 breaths/min, respectively), PEEP_i decreased (from 15.6 to 14.4 and 13.3 cm H₂O, respectively), and end-expiratory lung volume also fell. Increases in flow from 30 L/min to 60 and 90 L/min also led to decreases in the swings in P_{es} from 21.5 to 19.5 and 16.8 cm H₂O, respectively. (Used, with permission, from Laghi et al.¹¹⁰)

Mode-Specific Effects of Inspiratory Unloading

Pressure support and IMV are sometimes combined in a given patient. In an international survey of mechanical ventilation,¹¹³ this combination tied with assist-control ventilation as the most commonly used mode of ventilation in North America (34% for each). The rationale for combining the two modes is unclear. Presumably, clinicians use pressure support to overcome the work imposed by the endotracheal tube and demand valve during the non-mandatory breaths.

Examining the response of the respiratory centers to this combination of modes provides useful insight into patient-ventilator interaction. A decrease in the number of mandatory breaths produces a decrease in the average $\dot{V}_T$,⁹⁹ with inevitable increase in the ratio of dead space to $\dot{V}_T$. To avoid a decrease in alveolar ventilation, the patients increased respiratory motor output, inspiratory effort, and rate. Adding pressure support of 10 cm H₂O caused a decrease in effort at any given IMV rate. The decrease in effort during the mandatory ventilator breaths was related to the decrease in respiratory motor output during the intervening breaths ($r = 0.67$) (Fig. 53-20).⁹⁹ In other words, the reduction in motor output during the intervening breaths achieved by adding pressure support was carried over to the mandatory breaths, facilitating greater unloading. Combining IMV and pressure support provides a sometimes useful means of achieving a high level of assistance; the combination has a clinical advantage when it is difficult to achieve a high inspiratory flow in the assist-control mode, as with the Siemens 900C ventilator (Siemens Corporation, New York, NY), although few of these machines are likely to be in current use.

Figure 53-20



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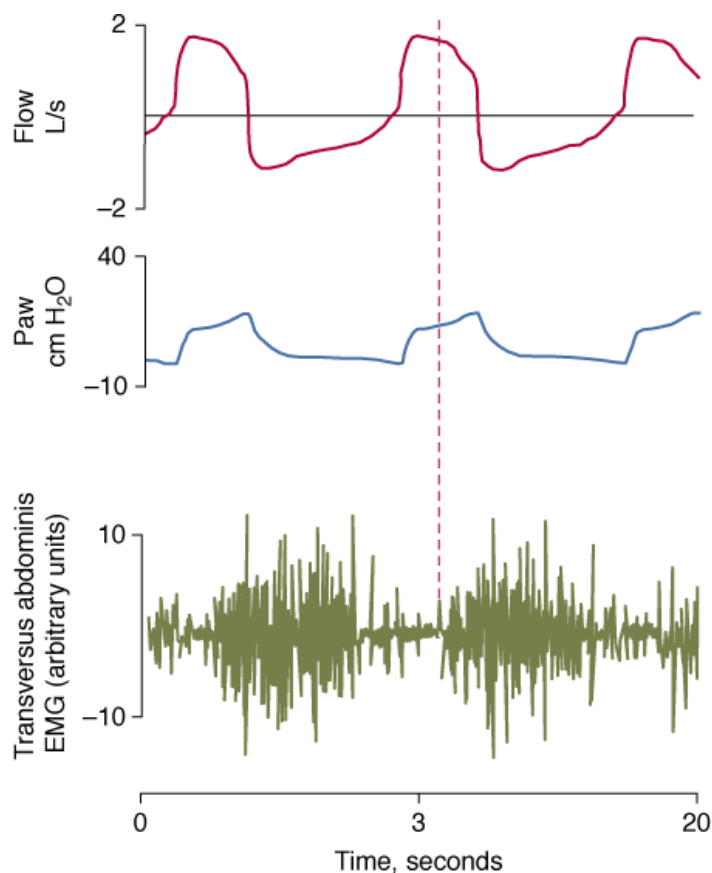
The change in pressure-time product per breath (PTP/br) during mandatory breaths (of IMV) consequent to the addition of pressure support of 10 $cm H_2O$ to a given level of IMV was related to the change in respiratory motor output (dP/dt) effected by pressure support during the intervening breaths ($r = 0.67$; $p < 0.0001$). The more that pressure support decreased respiratory motor output during the intervening breaths, the greater was the reduction in patient work during the mandatory ventilator breaths delivered during IMV. (Used, with permission, from Leung et al.⁹⁹)

Inspiration–Expiration Switching

In studies of interactions between patient effort and mechanical ventilation, remarkably little attention has been paid to the switch between inspiration and expiration. The most common mode of ventilation is some form of volume assistance,¹¹³ such as assist control or IMV. “Cycling-off” of mechanical inflation, however, may be based only indirectly on volume. Instead, inspiratory flow is commonly preset and the ventilator adjusts inspiratory time to achieve a given V_T . This system is more precisely termed *time-cycled ventilation*. Inflation time is constant with a time-cycled machine, but patients invariably display considerable breath-to-breath variability in inspiratory time.¹¹⁴ Accordingly, a patient’s neural inspiratory time may be shorter or longer than the inflation time of the machine. If the machine delivers the set V_T before the end of a patient’s neural inspiratory time, ventilator assistance will cease while the patient continues to make an inspiratory effort—with double triggering (two ventilator breaths for a single effort) a likely consequence.¹⁰⁴

During assist-control ventilation, when a patient’s neural inspiratory time is short, ventilator inflation may continue into neural expiration and thus decrease the time available for lung emptying. The sense of being unable to empty the lungs may cause patients to activate the expiratory muscles with the result that the patient appears to fight or buck the ventilator (Fig. 53-21). The decrease in time for emptying also increases the likelihood for dynamic hyperinflation and thus inspiratory efforts that fail to trigger the ventilator.

Figure 53-21



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Recordings of flow, airway pressure (P_{aw}), and transversus abdominis electromyography (EMG) in a critically ill patient with COPD receiving pressure support of 20 cm H₂O. The onset of expiratory muscle activity (*vertical dotted line*) occurred when mechanical inflation was only partly completed. (Used, with permission, from Parthasarathy et al.⁹⁷)

With pressure support, the algorithm used for “cycling-off” of mechanical inflation varies among brands (see [Chapter 8](#)). On most ventilators, the termination of inspiratory assistance during pressure support is set at a default threshold, such as 25% of the peak inspiratory flow or inspiratory flow of 5 L/min. That is, when inspiratory flow falls to 25% of the peak value or below 5 L/min, inspiratory pressurization switches off. Such algorithms can be problematic in patients with COPD because increases in resistance and compliance produce a slow time-constant of the respiratory system. The longer time needed for flow to fall to the threshold value can cause mechanical inflation to persist into neural expiration. In twelve patients with COPD receiving pressure support of 20 cm H₂O, five recruited their expiratory muscles while the machine was still inflating the thorax.¹¹⁵ The patients who recruited their expiratory muscles during mechanical inflation had an average time constant of 0.54 seconds, compared with an average of 0.38 seconds in the patients who did not exhibit expiratory muscle activity. The persistence of mechanical inflation into neural expiration is very uncomfortable, as is well recognized with use of inverse-ratio ventilation. Algorithms that achieve better coordination between the end of mechanical inflation and the onset of a patient’s expiration may lessen this form of patient-ventilator asynchrony.^{116,117}

Sleep–Wake State

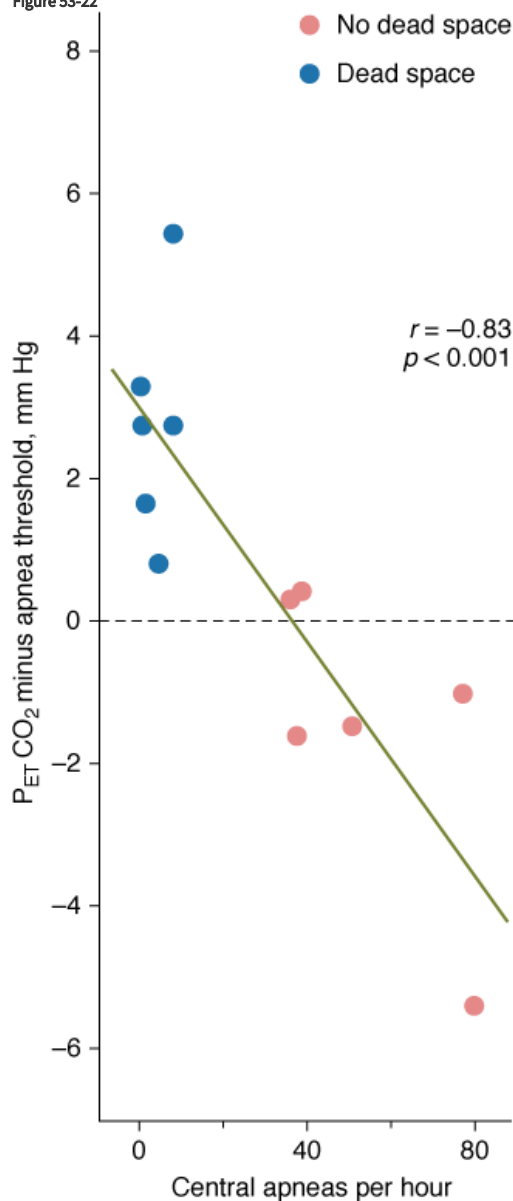
The transition between sleep and wakefulness can lead to respiratory distress in the ventilated patient. More surprisingly, the transition between wakefulness and sleep can also cause distress. Specifically, the selection of ventilator mode and settings can provoke sleep disruption. Ventilated patients experience considerable sleep disruption, with as many as seventy-nine arousals and awakenings per hour.^{118,119} Sleep disruption can adversely affect patient outcome.¹²⁰

In eleven critically ill patients, Parthasarathy and Tobin¹²¹ studied the interaction between ventilator mode and sleep. Sleep fragmentation was greater during pressure support than during assist-control ventilation: 79 versus 54 arousals and awakenings per hour (see [Fig. 53-9](#)). Six of the eleven patients developed central apneas during pressure support, but not during assist-control ventilation. V_T was 8 mL/kg during

assist-control; pressure support was titrated to achieve the same V_T . The level of pressure support was 16.8 ± 1.5 cm H₂O in patients with apneas and 19.6 ± 2.6 cm H₂O in patients without apneas; thus, apneas did not result simply from a higher level of pressure support.

Sleep fragmentation, measured as the sum of arousals and awakenings, was greater during pressure support than during assist-control: 79 ± 7 versus 54 ± 7 events per hour. Disturbed sleep during pressure support was related to the development of central apneas ($r = 0.57$), which, in turn, was significantly related to the difference between during resting breathing and the patient's apnea threshold (end-tidal CO₂ [$\Delta P_{ET}CO_2$]) ($r = -0.83$) (Fig. 53-22). $\Delta P_{ET}CO_2$ was the most important determinant for the development of apneas: As $\Delta P_{ET}CO_2$ grew wider, the number of central apneas increased. The addition of 100 mL of dead space to the ventilator circuit in the six patients who developed apneas produced a 4.3 mm Hg increase in end-tidal CO₂, decreased the frequency of central apneas, from fifty-three to four apneas per hour, and the frequency of arousals and awakenings, from eighty-three to forty-four events per hour. This study shows that while critically ill patients have a background level of sleep disturbance, secondary to factors such as pain, medications, staff interruptions, noise, and light, the mode of mechanical ventilation can further aggravate sleep disruption.

Figure 53-22



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The difference between the average end-tidal CO₂ ($P_{ET}CO_2$) and the apnea threshold plotted against the number of central apneas per hour of pressure support alone (red symbols) and pressure support with added dead space (blue symbols) in six patients. The average end-tidal CO₂ was measured during both sleep and wakefulness. The mean number of central apneas per hour was strongly correlated with the end-

tidal CO₂ during a mixture of both sleep and wakefulness (including the transitions between sleep and wakefulness) ($r = -0.83$, $p < 0.001$).

(Used, with permission, from Parthasarathy et al.¹²¹)

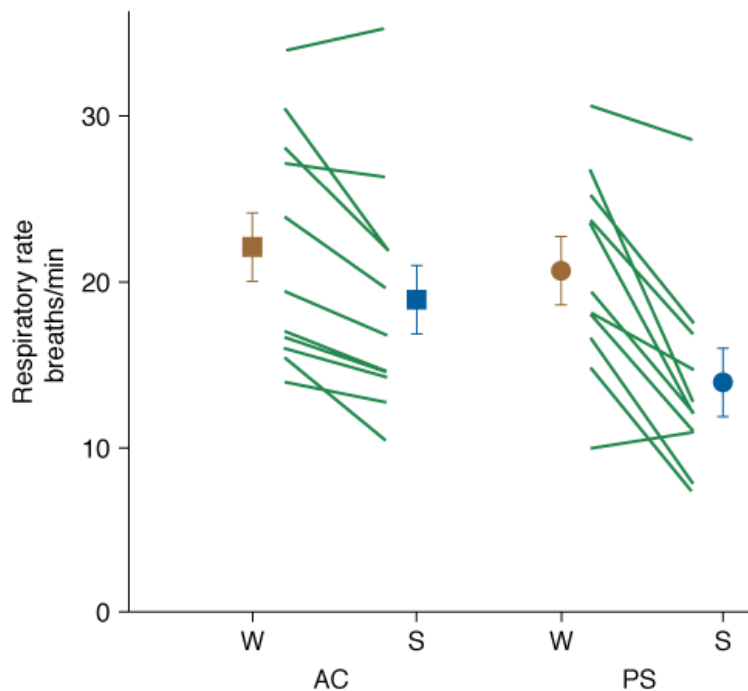
The observation that ventilator mode can aggravate sleep disruption was confirmed by Bosma et al.¹²² In thirteen patients undergoing polysomnography, these investigators adjusted the levels of pressure support ventilation and proportional-assist ventilation to achieve a similar degree of inspiratory muscle unloading. The number of arousals and awakenings were significantly greater with pressure support than with proportional-assist ventilation. Moreover, the number of patient-ventilator asynchronies per hour correlated significantly with the number of arousals per hour ($r^2 = 0.71$).

Cabello et al.¹²³ compared sleep quality while patients were randomized between assist-control ventilation, pressure support set by the patient's attending physician, and pressure support continuously adjusted by a closed-loop knowledge-based system. In contrast to the report by Parthasarathy and Tobin,¹²¹ sleep architecture, sleep quantity, and sleep fragmentation were equivalent with the three ventilator modes.

The different results in the two studies can be explained by the method of selecting tidal volume in the study of Cabello et al.¹²³ A tidal volume of 8 mL/kg was employed during assist-control ventilation, whereas a tidal volume between 6 and 8 mL/kg was targeted during pressure support.¹²³ The median (twenty-fifth to seventy-fifth percentile) tidal volumes were 500 mL (380 to 500 mL) during assist-control ventilation, 450 mL (357 to 521 mL) during clinician-adjusted pressure support, and 390 mL (330 to 492 mL) when a closed-loop knowledge-based system was used to continuously adjust pressure support. The findings of these three studies of mechanical ventilation during sleep in critically ill patients indicate that pressure support, when carefully adjusted to avoid hyperventilation, does not increase the amount of sleep fragmentation over that experienced with assist-control ventilation. If, however, pressure support is set in the manner of everyday clinical practice, it is likely to lead to sleep fragmentation.^{121,122}

The alterations in breathing pattern and gas exchange induced by sleep have important implications for the selection of ventilator settings. During pressure support, sleep induced a 23% increase in inspiratory time and a 126% increase in expiratory time, as compared with wakefulness.¹²¹ Sleep caused the respiratory rate to decrease by 33% during pressure support and by 15% during assist-control ventilation (Fig. 53-23). The level of pressure support is commonly titrated to respiratory rate, which provides a reasonable guide to patient effort.^{115,124} Pressure support is commonly titrated during the daytime without the clinician being sure whether the patient is asleep or awake. If the patient is asleep at the time of the adjustment, a point at which respiratory rate will be relatively low, then, on awakening, the increase in a patient's rate may cause a considerable increase in effort.

Figure 53-23



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Respiratory rate during assist-control ventilation (AC) and pressure support (PS) in eleven critically ill patients. For each mode, the lines connect the mean value for each patient during wakefulness (W, left) and sleep (S, right). Compared with wakefulness, group mean respiratory rate was lower during sleep (blue symbols) than during wakefulness (red symbols). The difference between sleep and wakefulness was greater for pressure support than for assist-control ventilation. (Modified, with permission, from Parthasarathy and Tobin.¹²¹)

Pharmacotherapy

If a specific cause of acute distress cannot be detected and corrected (see Table 53-1) and reassurance provides no relief, pharmacologic agents are commonly employed. The agents most often used are derived from the following primary classes: opiate analgesics, benzodiazepines, selective α_2 agonist,¹²⁵ and neuromuscular blocking agents. Chapter 50 provides a complete discussion of these agents.

In managing the patient who is fighting the ventilator, neuromuscular blocking agents have several disadvantages: They mask a patient's complaints and physical findings; unrecognized disconnection of the ventilator circuit can produce apnea with catastrophic hypoxemia; elimination of cough predisposes to the development of atelectasis; and prolonged paralysis or weakness may persist after their use.

Summary

Sudden distress in a ventilator-supported patient is a medical emergency. The first rule is to ensure adequate ventilation. The patient should be disconnected from the ventilator and manually ventilated with 100% oxygen. While this is being performed, a systematic effort should be made to try to determine the cause of distress and correct it. If the distress is the result of poor coordination of a patient's respiratory efforts with the rhythm of the ventilator, this can usually be resolved by careful adjustment of the ventilator settings and the administration of analgesic or sedative agents, or both.

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